



CLIMATE VULNERABILITY ASSESSMENT

SERBIA





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SERBIA

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INTRODUCTION

Combating climate change represents today one of the key issues on the international policy agenda. Climate change has been already affecting ecosystems in a negative way and these changes will increase in the rate of their occurrence as well as in the degree of their effects during this century. This means that food and water will be less available, natural disasters more often, people's health at risk, some species will disappear, and ecosystems will be destroyed or degraded. The expected effects of climate change are complex and far-reaching, so that it is not surprising that numerous global initiatives have focused precisely on these issues. Apart from the activities to mitigate climate change, urgent development of adaptation systems to climate change has become inevitable. The correlation between adaptation to climate change and other developmental problems as well as the necessity of an integrated approach in finding solutions pose a special challenge in all sectors, including politics, science, economy and civil society.

The European Union has developed a set of strategic and legal papers related to climate change. The EU White Paper on Adaptation to Climate Change refers most directly to this issue. With this paper, the EU has set a framework to tackle adaptation problems at the Union's level, concurrently committing its member states to approach seriously and strategically to finding solutions to this problem. As an EU membership candidate country, Serbia will soon face these commitments and the issue of climate change, particularly adaptation to it, will certainly be intensively dealt with in the near future. Earlier activities focusing on climate change adaptation in Serbia have been very moderate and we hope that this paper will contribute to a faster and more appropriate way of setting up a climate change adaptation system.

The "South East European Forum on Climate Change Adaptation" has been founded within a project bearing the same name and has gathered around 80 members organized in four national networks in a region that has been recognized as one of the areas of Europe most vulnerable to climate change. Four national civil society networks have been established in Serbia, Croatia, Macedonia and Montenegro with the aim to strengthen the capacities in the civil sector and to raise public awareness and enhance the dialogue with the decision-

makers by organizing activities, at the national as well as at the regional level, in order to draw attention to climate change issues. This report on climate change vulnerability assessment has been elaborated within the above mentioned project.

The recommendations given in this paper are meant to serve for improvement, better defining and cooperation of all initiatives in the field of climate change as well as to direct the activities of the "Climate Forum", consists of 17 civil society organizations in Serbia.

Climate change has a significant negative socio-economic impact, and one of the key objectives of the project has been the enhancement of sustainable management of natural resources, with the focus on the functionality of different ecosystems. This overall threat demands a flexible managing approach that enables inclusion of all relevant sectors, foremost the sectors of agriculture, water management, forestry, nature protection, public health and energy. Scientific data stress the need to act immediately, as the extreme weather events, climate change-induced droughts and heat waves, will become even more intensive in the future.

The project „South East European Forum on Climate Change Adaptation" has been implemented by the Red Cross of Austria in cooperation with the national Red Cross Societies in Montenegro, Croatia and Macedonia, as well as with the Environmental Improvement Center from Serbia. Regional project partners are WWF Danube-Carpathian Programme, the Institute for Economic Promotion of the Austrian Federal Economic Chamber, Red Cross/Red Crescent Climate Center.

We would like to thank all members of the "Climate Forum" who contributed to "Climate Vulnerability Assessment". Special thanks are also due to my colleagues who provided support and advice during the writing-up process: Sonja Badjura, Milena Miladinović, Ivana Jovčić and Ana Anđelković. The project has been funded by the European Union through the Instrument for Pre-Accession Assistance (IPA).

Further information available at:
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LIST OF ACRONYMS

CCA	Climate Change Adaptation
CSO	Civil Society Organisation
CVA	Climate Vulnerability Assessment
GDP	Gross Domestic Product
DRR	Disaster Risk Reduction
EC	European Commission
EIC	Environmental Improvement Center
EU	European Union
EWS	Early Warning System
FYR	Macedonia Former Yugoslav Republic of Macedonia
IPA	Instrument for Pre-accession Assistance
IPCC	Intergovernmental Panel on Climate Change
NGO	Non-Governmental Organisation
RCCC	Red Cross/Red Crescent Climate Centre
RC/RC	Red Cross/Red Crescent
SEE/CCFAP-A	South East European Climate Change Framework Action Plan for Adaptation
SEEFCCA	South East European Forum on Climate Change Adaptation
SEEVCCC	South East European Virtual Climate Change Centre
UN	United Nations
UNDP	United Nations Development Programme
UNFCCC	United Nations Framework Convention on Climate Change
UNISDR	United Nations International Strategy on Disaster Reduction
WHO	World Health Organization
WWF	World Wide Fund for Nature

SCOPE AND PURPOSE OF THE STUDY

Southeast Europe has been recognized as one of the regions vulnerable to climate change. So far, several regional initiatives have been launched with the aim to strengthen capacities in combating climate change in this part of Europe. One of them is also the project South East European Forum on Climate Change Adaptation, within which this analysis has been elaborated, financed by the EU through IPA funds. The analysis has focused on adaptation to climate change for several reasons. First of all, it was apparent that mitigation measures would not be sufficient and that climate change could not be stopped over a short period of time. Therefore, it is necessary to develop and implement adaptive measures in order to alleviate the effects of climate change. As adaptations are slowly taking place among priorities in global politics, there is a need in all countries to elaborate vulnerability evaluation in terms of climate change effects and appropriate adaptation plans. Such objectives have also been set by the European Union in their strategic documents on climate change. As Southeast European countries are clearly oriented

towards the EU membership, they will soon have to pay special attention to elaboration of appropriate papers and development of their adaptive capacity.

The vulnerability assessment to climate change and adaptation planning are very complex processes that require analyses in all sectors of society and involvement of numerous experts and other stakeholders. Bearing this in mind, this Climate Vulnerability Analysis (CVA) has no pretension to be a national umbrella paper that would give final vulnerability evaluation and set basic strategic directions in the development of adaptation. Instead, it aims at initiating the process of adaptation planning and in-depth discussions, as well as to stress the importance and urgency of these processes by awareness-raising, advocacy and lobbying. The analysis is particularly directed towards the civil sector that would have to play a very important role in the process of adaptation planning. It is also partly designed to help civil society organizations streamline their activities in the field of climate change.

METHODOLOGY

The Climate Vulnerability Analysis (CVA) is based on the analysis of the existing global and country-specific data. For the purposes of this analysis, the following sources have been used:

- scientific papers, publications and projects;
- international, regional and national legal and strategic documents;
- reports published by state authorities and public institutions;
- publications, projects and initiatives of the civil sector;
- statistical data that are important for individual sectors;

In addition to this, experts from the relevant fields and members of the South East European Forum on Climate Change have also been consulted.

Apart from the general analysis of the climate change impacts and vulnerability on the national level, five sectors have been analyzed in depth. These are:

water resources, public health, agriculture, energy and biodiversity. Member organizations of the "South East European Forum for Climate Change Adaptation" have participated in the designation of these sectors. A brief introduction concerning the current situation is provided for each sector, describing the scope and manner of climate change impact on each of the sectors in terms of international experience and existing national data, concluding with a set of recommendations for potential adaptation measures.

The scope and structure of the analysis has been harmonized with the project teams in partner countries of the IPA project to ensure compatibility of the analyses as well as comparison and synthesis of regional reports on vulnerability to climate change.

The content drafts have been presented to member organizations of the Forum during three workshops organized within the IPA project in 2012 (in Zagreb, Belgrade and Sutomore).

EXECUTIVE SUMMARY

This study represents a climate change vulnerability analysis of Serbia, predominantly focusing on adaptation measures, i.e. on development possibilities of adaptation systems to climate change. The study has been elaborated within the IPA project “South East European Forum on Climate Change Adaptation” and as such it does not aim at becoming an umbrella national paper in the field of adaptation. It is more focused on the civil sector, i.e. on initiating the adaptation process planning and public discussions about climate change as well as awareness-raising about all issues related to climate change.

According to most forecasts, significant climate change is expected in Serbia and the whole region of Southeast Europe in the near future. Projections of regional climate models predict that by the end of this century the annual average temperatures will rise from 2.4°C to 2.8°C under an optimistic scenario (A1B1), respectively from 3.4°C to 3.8°C under a pessimistic scenario (A2). Under all scenarios a raise of average temperature is expected in all parts of Serbia, with some regional differences. The rainfall situation is somewhat more complex. Under the A1B1 scenario, a reduction of precipitation is expected throughout Serbia, with a little less pronounced decrease in Vojvodina and some parts of eastern Serbia. Under the A2 scenario, the precipitation will increase in Vojvodina, while in other parts of Serbia it will remain the same or decrease.

It is very likely that in the near future Serbia will be also faced with an increased number of natural disasters, primarily floods, droughts and fires.

The Serbian economy relies to a great extent on activities that are dependent on climate conditions and thus vulnerable to climate change. Problems that may occur due to climate change and their effects on economy and the society as a whole have not been thoroughly considered. Also, Serbia has still no national strategy on climate change adaptation; moreover, the vast majority of sectoral strategies and plans do not consider any adaptation measures to climate change. Taking all this into account, we can say that the current adaptive capacity in Serbia is at a low level. Some institutions and organizations have lately made substantial progress in building their

capacities in fighting climate change, but there is still no synchronized and clearly defined climate change adaptation policy at the national level.

Five sectors have been further analyzed in the study: water management, public health, agriculture, energy and biodiversity. In each of them, a substantial exposure and vulnerability to climate change has been identified. Furthermore, each sector is faced with numerous problems that do not correlate directly to climatic characteristics, but they reduce their adaptive capacity, i.e. their adaptation possibilities to climate change. Climate change can also enhance new development possibilities, especially in agriculture, but in order to make use of them, the individual producers need a strategic approach and system support.

Problems that pervade all sectors and represent at the same time also the main obstacles for the development of an efficient adaptation system are disparities in the regional development, depopulation of rural areas, centralization of most sectoral systems, underdeveloped infrastructure, obsolete technology, lack of strategic approach in planning, lack of an institutional organization dealing with climate change issues, insufficient intersectoral cooperation and generally a low public awareness related to climate change.

In terms of adaptation measures, specific recommendations have been made for each sector. In accordance with the identified pervading problems, main recommendations include: elaboration of strategic adaptation papers, establishing of coordination bodies for planning and implementation of adaptation measures, systemic development of intersectoral cooperation, strengthening of scientific capacities and development of cooperation between the scientific sector and decision-makers, and inclusion of the civil society in the adaptation planning process.

It has been estimated that climate change will mostly affect the population living in remote, underdeveloped areas. This population relies predominately on agriculture, and their adaptive capacity is at an extremely low level due to depopulation and lack of economic development in certain areas.

The civil society organizations play an important role in addressing climate change. Their work is especially relevant at this time when the awareness of the problems that may arise due to climate change is low. Main recommendations for such organizations are inclusion in planning processes at the national level, building their

capacity through networking, establishing collaboration and project development with local population and especially vulnerable groups, specialization for particular adaptation problems and going into partnership with the scientific sector, and development of regional and international cooperation.

1 Introduction: climate change

1.1. THE SCIENTIFIC BASIS: CLIMATE CHANGE AS A RESULT OF THE GREENHOUSE EFFECT

1.1.1. Definition of climate change

“Climate change” means a change of climate which is attributed directly or indirectly to human activity that alters the composition of the global atmosphere and which is in addition to natural climate variability observed over comparable time periods.” (UNFCCC, 1992: 3)

In the broadest sense, climate change is a consequence of complex abiotic and biotic processes, reflected by statistically significant changes of climate parameters over a longer period of time. Factors that initiate climate change could be of astronomic, geophysical or biotic nature. Astronomical and geophysical factors are external forcers of climate change, as they evolve outside the atmosphere. Astronomical factors refer to the activities of other astronomical objects, notably the Sun, or to the interaction of these objects and the Earth (in terms of distance, trajectory, respective positions, inclination, etc.). The geophysical factors relate to the Earth's tectonic activity. The effects of these processes, such as volcanic eruptions, tectonic movements or inclination changes, can have a direct impact on the climate.

Biotic factors can be further divided into anthropogenic and non-anthropogenic ones. The ecosystem processes are the essence of non-anthropogenic factors such as primary production (photosynthesis) and the water and nutrient cycle.

Anthropogenic factors are human activities which affect the climate. Nowadays, the understanding of *climate change* is generally associated with human-induced climate change. It can be said that in this narrower sense, *climate change* is a change of climate resulting from human activities in the biosphere.

The human impact on climate has been disregarded for a long time. The first report of the Intergovernmental Panel on Climate Change (IPCC) was completed in 1990, leading to an almost global consensus on human impact on climate. At the same time, it served as the basis of the United Nations Framework Convention on Climate Change.

The main conclusions of this report are as follows:

- Greenhouse effect exists – emissions induced by human activities are significantly increasing atmospheric concentrations of greenhouse gases (carbon dioxide, methane, Freon, nitrous oxide, etc.). These increases will enhance the greenhouse effect, resulting in atmospheric warming;
- Emissions of carbon dioxide are responsible for more than a half of the greenhouse effect increase;
- During the 21st century, global mean temperature is expected to increase by 0.2°C per decade;
- there are many uncertainties in predictions and projections especially in terms of timing, intensity and regional specificities of climate change;
- Due to thermal expansion and the melting of ice expected sea level rise is 6 cm per year.

1.1.2. The greenhouse effect mechanism

The greenhouse effect involves the absorption of thermal radiation by greenhouse gases and its re-radiation in all directions. Since some of this infrared radiation returns to the Earth's surface, it therefore further warms the surface. This in turn increases the emission of infrared radiation, so that this effect essentially represents a positive feedback loop.

Greenhouse gases are atmospheric gases that absorb and emit long-wave (infrared) radiation. The most

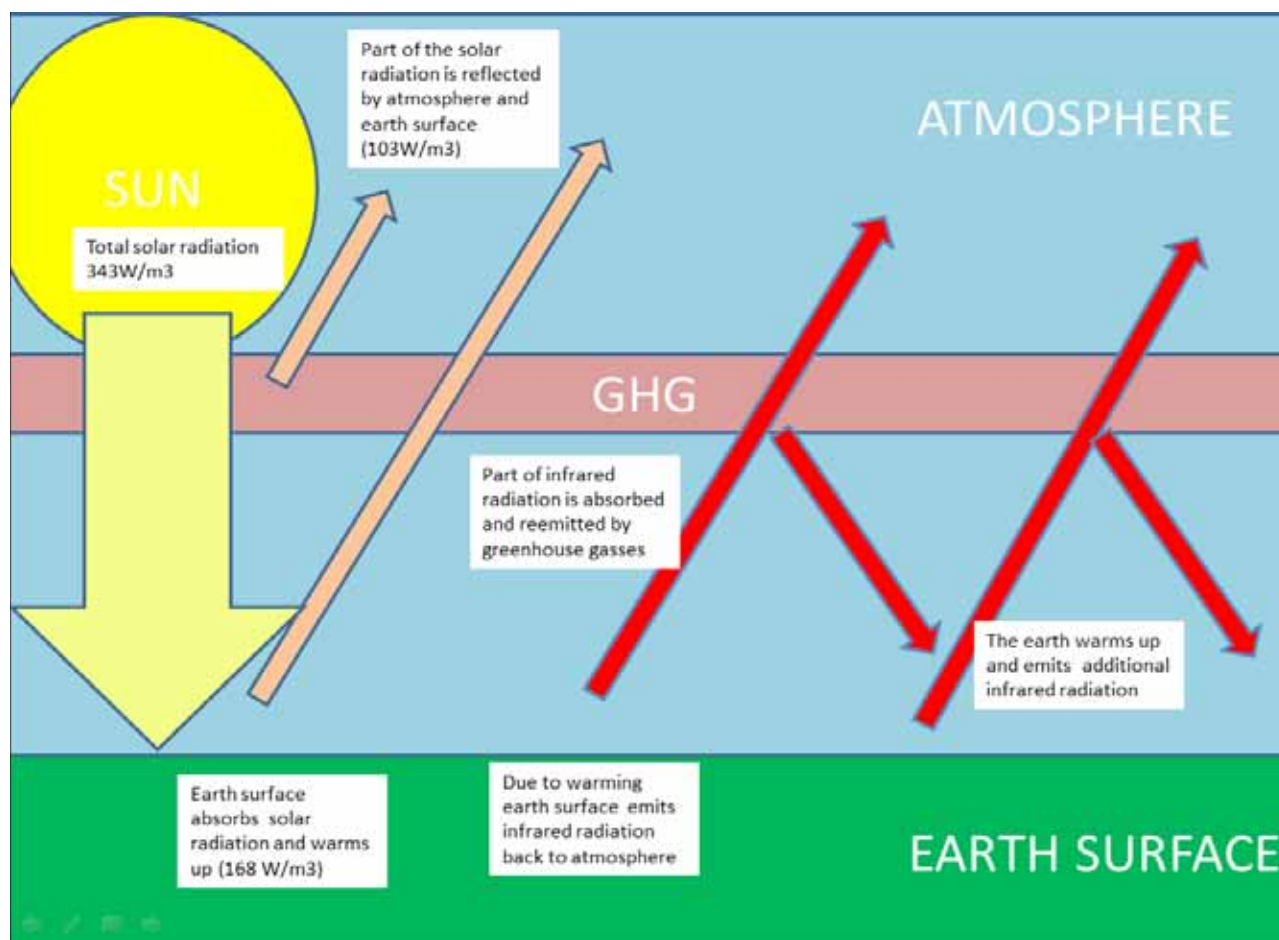
common greenhouse gases are: water vapor, carbon dioxide, methane, nitrous oxide and ozone. There are also a number of other gases in smaller concentration in the atmosphere, such as CFC (CClF_2), HCFC (CHClF_2), tetrafluoromethane (CF_4), hexafluorethane (C_2F_6), sulfur hexafluoride (SF_6), and nitrogen trifluoride (NF_3).

The most important initiator of global warming is carbon dioxide. The concentration of this gas in the atmosphere largely depends on human activities, i.e. the increase of carbon dioxide concentration in the atmosphere results from emissions of this gas that are consequences of burning fossil fuels and deforestation.

Over the past several decades, the atmospheric concentration of carbon dioxide has significantly increased and has been directly linked to the observed rise in global temperature levels.

1.1.3. Climate change effects

The gradual warming of the atmosphere has numerous and far-reaching consequences for the entire world. Despite the intense technological development, human civilization is essentially still tied to natural systems and depends directly on the processes taking place within these systems. Direct effects of climate change, including higher temperatures, melting ice caps and rising sea and ocean levels, changing precipitation regimes, may significantly disrupt the functioning of human society. The production and accessibility of food and, water, public health, transport and energy supply, are just a few of the key elements for a functioning human community that are highly dependant on climatic conditions and that could be significantly destabilized by climate change. Increasingly frequent catastrophes such as floods, droughts and fires are also related to climate change. With medium level reliability, it can be asserted that the frequency of droughts and floods has increased in certain parts of the world and that this trend is linked



III. 1. Greenhouse mechanism as per Houghton et al (1996)

with the climate change¹. The world will very likely face even greater risks of such disasters in the future.

By the end of the 21st century, the Fourth Report of the Intergovernmental Panel on Climate Change², and

¹ IPCC, 2012

² IPCC, 2007

the most optimistic scenario estimates temperature increase of 1.8°C, while the most pessimistic scenario foresees a rise of 4°C. The two most extreme scenarios predict the sea level rise of 18-38 cm and 26-59 cm respectively³.

³ Ibid.

1.2. CLIMATE CHANGE MITIGATION EFFORTS THROUGH REDUCTION OF GREENHOUSE GASE EMISSIONS

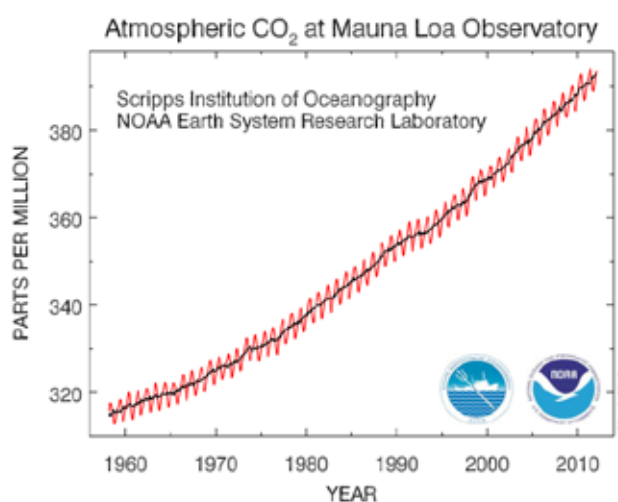
“Mitigation is an anthropogenic intervention to reduce the anthropogenic forcing of the climate system; it includes strategies to reduce greenhouse gas sources and enhancing greenhouse gas sinks.” (IPCC, 2007)

Atmospheric concentrations of carbon dioxide, the most significant cause of the greenhouse effect, have increased by 35% (100ppm) over the past 250 years. In the space of only ten years, from 1995 to 2005, the concentration rose by 19 ppm, which was the highest growth rate ever recorded. Atmospheric concentrations of carbon dioxide have been caused primarily by anthropogenic emissions deriving from fossil fuel use, land-use change and biomass combustion. Fossil fuel emissions have grown by 70% over the past thirty years⁴. During the 1990s, the annual *carbon release* from fossil fuel *combustion* was estimated to be 6 Gt, which increased to 7.8 Gt in the early 21st century already. Emissions due to land-use change, i.e. deforestation contribute about 12% of total anthropogenic carbon dioxide emissions.

Such a rapid growth of anthropogenic carbon dioxide emissions led to global consensus on the need to reduce emissions in order to prevent and mitigate climate change. Greenhouse gas emissions reduction and the fight against climate change have become priority issues for many countries and international organizations. The main document serving as a guideline for implementation and coordination of international efforts to reduce the carbon dioxide emissions is a treaty entitled the United Nations Framework Convention on Climate Change, produced at the United Nations Conference on Environment and Development, informally known as the Earth Summit, held in Rio de Janeiro in 1992. The treaty itself is not legally binding.

⁴ Marland et al., 2006

However, it envisages the adoption of protocols that set out provisions binding on signatories. One of the most important of these protocols is the Kyoto Protocol, initially adopted in 1997, and entered into force in 2005. Under the Protocol, 37 most developed signatories (Annex I countries) have the obligation to reduce their emissions to 5% below 1990 levels by 2008-2012. Aside from these stipulations, the Protocol allows for several flexible mechanisms, such as the International Emissions Trading, Clean Development Mechanism and Joint Implementation.



III.2. Upward trend in atmospheric carbon dioxide concentrations as measured at the Mauna Loa Observatory. Retrieved from: www.esrl.noaa.gov

Apart from this global initiative, different regional initiatives exist as well, of which, as far as Serbia is concerned, the most relevant one is the European Union initiative, known as the “20-20-20” Targets. Namely, the European Union has adopted a set of legislative acts known as the Climate and Energy Package,⁵

⁵ http://ec.europa.eu/clima/policies/package/index_en.htm

binding member states to reduce their emissions by 20% below 1990 levels by 2020; increase the share of renewable energy in electricity generation by 20%; and a 20% cut in primary energy consumption through efficiency improvements.

All measures based on emissions reduction, i.e. prevention of climate change causes are called mitigations or mitigation measures. In spite of all efforts to reduce the emissions as one of the main causes of climate change, it is evident that sole implementation of mitigation measures is not enough to withstand climate change. The effects of climate change are already visible and clearly, they will become more and more severe in the near future.



III.3. Glacier melting in the Austrian Alps. Picture: G.Sekulic

The need for adaptation “Adaptation is the adjustment of natural or human systems as a response to actual or expected climate stimuli or their effects, which moderates harm or exploits beneficial opportunities.” (IPCC, 2007.)

Even if there greenhouse gas emissions were drastically reduced now, the effects of emissions that occurred up to now would be still felt for years to come. Basically, climate change cannot be stopped in the short term. The prognosis indicates that the greenhouse gases emissions will continue to grow over the next few decades, and it is impossible to expect positive effects of the implemented mitigation measures for years⁶. For that reason, the international community has lately paid special attention to adaptation measures, i.e. measures to adjust to climate change. Adaptation involves finding ways to reduce the vulnerability of natural and human systems against climate change

effects. Essentially, adaptation represents greater resilience to climate change.

The fact that all predictions are subject to many uncertainties should be borne in mind in all analyses of possible adaptation measures to climate change.

These uncertainties result from the use of different prediction methods (different models, choice of scenarios, choice of time periods for verification or prediction, selection of methods for breaking down the model results from global to regional levels, etc.), as well as from the unreliability of used data (insufficient number and quality of certain data, incompatibility of different research methods, etc.) However, it must also be borne in mind that these uncertainties are not of the same order of magnitude. For instance, the prediction that the average and seasonal temperature will rise is quite uncertain, contrary to the prediction that the annual precipitation will decrease, while types of changes that may occur at the seasonal level are especially unpredictable.

Given the many uncertainties, the potential adaptation measures to climate change should not cause adverse effects in case the predicted changes do not occur. In other words, it is desirable to implement measures that would not only reduce the risk of climate change, but also have other positive effects on society, economy and environment.

⁶ IPCC, 2007

2 Current and future climate risks in Serbia

2.1 SERBIA: GENERAL CHARACTERISTICS

2.1.1 Geographical features

Serbia, a landlocked country located in southeastern Europe, occupies an area of 88,361 km². A large part of its territory (80%) lies in the Balkans, while the remaining 20% is situated in the Pannonian Plain.

In terms of biogeography, Serbia is divided into the Pannonian, Continental and Alpine regions (Dinaric Alps). The northern part of the country (Vojvodina) is a plain, while the rest of the terrain mainly consists of hills and mountains dissected by river valleys. The altitude ranges from 17 meters above sea level along the Danube River in Prahovo, to 2656 meters, which is the altitude of the highest mountain peak Djeravica, located in Kosovo. Large rivers include the Danube, Sava, Drina, Velika Morava, Tisa, Tamis, Juzna Morava, Zapadna Morava, Ibar, Lim, and Timok. The Danube drainage basin covers 92.5% of the waters, 5.5% belongs to the Adriatic drainage basin, and about 2% to the Aegean basin. The largest lakes are reservoirs (Iron Gate Reservoir, Vlasina Lake, Perucac Lake, Lake Celije and Gruza Lake). The largest natural lake is Belo Jezero, nowadays transformed into a fish pond.

Approximately one third of the territory (2.5 x 10⁶ ha) is forested. In Vojvodina, though, the level of afforestation amounts to only 6.5%. Deciduous forests prevail throughout Serbia. A much smaller area is covered by coniferous forests, making up only about 10% of the forested area.

Serbia's total border length amounts to 2,397 km. The country borders Croatia, Hungary, Romania, Bulgaria, Macedonia, Albania, Montenegro, and Bosnia and Herzegovina. Serbia has two autonomous provinces, Vojvodina and Kosovo and Metohija. The

whole state is divided into 29 administrative districts and 150 municipalities.

2.1.2 Demographics

According to the 2011 census⁷, Serbia excluding Kosovo has a total population of 7,120,666. The population is unevenly distributed and about 50% of the total population lives in Belgrade and in Vojvodina. Serbia has a negative demographic development and compared to the data of the 2002 census a 5% drop has been recorded.

Population growth has been recorded only in 22 municipalities, while all others have shown a downward trend in the last census period. There are 2,497,187 households in Serbia. This number has also been reduced, compared to the data from the last census period.

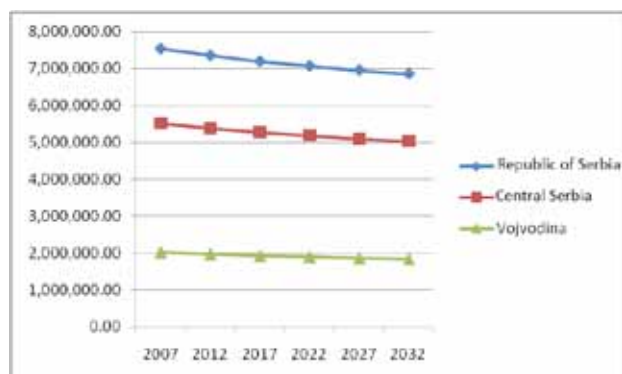
According to the data reported in 2010, the average age of the population in the Republic of Serbia is 41.3 years. The population aging index is 111.4; while life expectancy at birth is 74.0 years. Serbia has a very high proportion of population older than 65 years (approx. 17%), which ranks it among the countries that are regarded as old.

2.1.3 General economic indicators

The rate of economic growth in Serbia in the year 2011 amounted to 2%, after having achieved a modest growth of 1% in the previous year, 2010. Further slowdown of economic activity and the expansion in the Euro zone crisis had led to the correction of the projections of the economic growth in Serbia in the year 2012 to 1.5%, although it was originally expected to be at the last

7 Kosovo (UN 1244) has not been included in this census

year's level. Gross domestic product of Serbia in the year 2011 amounted to EUR 32.9 billion, which is an increase of 13.67% compared to 2010, when it was EUR 29 billion.



III.4. Population growth projections – source: Statistical Office of the Republic of Serbia, 2012 (<http://webzr.stat.gov.rs>)

The total foreign trade exchange of Serbia in 2011 amounted to EUR 22.88 billion. Exports amounted to EUR 8.43 billion, while imports amounted to EUR 14.44 billion. The import - export ratio was 58.4% and recorded a trade deficit of EUR 6 billion, while a surplus of EUR 1.07 billion has been achieved in the trade with the CEFTA⁸ countries. Exports of domestic products to the EU amounted to EUR 4.86 billion. Foreign direct investments in Serbia have doubled compared to 2010 and reached an amount of EUR 1.5 billion. Bearing in mind that even 57.67% of total exports from Serbia go to the EU countries, the economic slowdown in the Euro zone and the general decline in capital inflows from abroad will continue to be reflected in the production and export of Serbia.

With a GDP of EUR 4543.00 per capita, Serbia holds the 87th place in the world and measured by the Human Development Index (HDI) the situation is somewhat better, since Serbia is ranked 59 out of 187 countries. Together with its neighbors, Serbia belongs to the second group consisting of countries with a high level of human development. The unemployment rate

⁸ The Central European Free Trade Agreement signed in 2006 between Albania, Bosnia and Herzegovina, Croatia, Macedonia, Moldova, Montenegro, Serbia and the United Nations Interim Administration Mission in Kosovo (UNMIK)

Age group	%
0-14 years	15,1
0-19 years	20,9
20-39 years	27,2
40-59 years	28,5
60 and above	23,3
65 and above	16,9

Table 1: Serbian population by age groups. Retrieved from: <http://webzr.stat.gov.rs>, 2012

continues to be extremely high and in the period of the last eight years, it has reached a record level of 24.4% in 2011, with a tendency of further growth. According to the Statistical Office, the number of employed persons was 1.746 million last year, while the average net salary was RSD 37,976.00 (EUR 366.76).

The inflation reached the level of 14.5% in April last year, but went down to one digit number of 7% by December. The inflation rate projection for 2012 was 4%. At the same time, the Serbian currency has been constantly losing value and the official middle exchange course of Dinar to Euro was 109.392 RSD on 24 February 2012. Serbia's public debt at the end of the year 2011 exceeded over the legal limits and amounted to 46.4% of GDP, while the fiscal deficit amounted to RSD 5 billion, which makes it further complicated to achieve the set deficit target of 4.25% of GDP in 2012.

According to the latest data compiled by the Ministry of Labor and Social Policy, 700.000 persons lived below the poverty line in Serbia in 2011, which means that more than half a million people had an income lower than 8.500 dinars per month.

With 86.000 refugees and 210.000 internally displaced persons, Serbia is still ranked first in Europe in terms of forced migration⁹. In 2011, there were still 60 collective centers that housed more 4.700 refugees and internally displaced persons¹⁰.

⁹ UNHCR Global Report 2010. <http://www.unhcr.org/gr10/index.html#/home>

¹⁰ <http://www.srbija.gov.rs/pages/article.php?id=45>

SECTOR	No. of employees
A Agriculture, forestry and fishing	35484
B Mining	21554
C Manufacturing industry	296398
D Supply of electricity, gas, steam and air conditioning	27977
E Water supply, wastewater management, controlling of waste removal process, and similar activities	32218
F Building industry	72747
G Wholesale and retail trade; repair of motor vehicles and motorcycles	185515
H Traffic and warehousing	86439
I Accommodation and catering services	20511
J Information and communication	37260
K Finance and insurance	38877
L Real estate	3166
M Experts, scientific and technical sector	53272
N Administration and support services	30983
O Public administration and defense; social security service	70044
P Education	139640
Q Health and social insurance services	160620
R Art; entertainment and recreation	22315
S Other services	14195
TOTAL	1.349.215

Table 2. Number of employees in legal entities in Serbia by sectors. Retrieved from: <http://webzrzs.stat.gov.rs>, 2012.

2.2 CLIMATE

2.2.1 Current climate situation

The climate in Serbia is moderate continental with diverse local particularities. Factors that affect the climate on a local level depend on geographic position, relief and local influences (terrain exposure, presence of river systems, vegetation, urbanization, etc.). Geographic areas that significantly affect the climate of Serbia are the Alps, Mediterranean Sea, Gulf of Genoa, Pannonian Plain, Morava River Valley, Carpathians, Rhodopes and other mountainous areas with their valleys and plateaus. The position of major river valleys in the north-south direction allows deep penetration of northern air masses to the south.

Mean annual temperatures range from 3°C in the highlands over 1,500 meters above sea level to 12°C in lowland areas.

The mean annual air temperature in the period from 1961 to 1990 in areas of altitude up to 300 meters above sea level was 10.9°C. In areas of altitude from 300 to 500 meters mean annual temperature was approx. 10.0°C, and over 1000 meters the temperature was 6.0°C.

The absolute maximum temperature in the period of 1961 to 1990 was measured in the month of July, and it ranged from 37.1 do 42.3°C in lowlands, whereas in the mountainous areas it ranged from 27.6 to 34.0°C. The absolute minimum temperatures were registered in the month of January, and ranged from -30.7 to -21.0°C in lowland areas and -35.6 do -20.6°C in the mountainous regions.

Annual precipitation increases on average with the altitude. In lower areas the annual rainfall varies in the range from 540 to 820 mm. Areas of altitude above 1000 m have on average 700 to 1000 mm rainfall, whereas some mountain peaks in southwest Serbia have heavier precipitation up to 1500 mm. The larger part of Serbia has a continental precipitation regime, with heavier rainfall in the warmer season, which is not the case in some southwest parts of the country where the heaviest rainfall has been observed in autumn. June is the rainiest month, with on average 12 to 13% of total annual precipitation. The driest months are February and October. The occurrence of snow cover is characteristic for the colder part of the year; a maximum of snowy days has been recorded in January. The amount of solar radiation ranges in the interval from 1500 to 2200 hours per year. Surface air circulation is largely caused by orography. Winds from the northwest and west prevail in the warmer time of the year. During the colder period of the year predominantly blow winds from east and southeast, the latter called Kosava. Winds from the southwest prevail in the mountainous areas in southwestern Serbia¹¹.

2.2.2 Climate projections

According to the fourth report of the Intergovernmental Panel on Climate Change (IPCC), Europe is marked as one of the regions that are particularly vulnerable to climate change. Mean annual temperatures have risen faster in Europe than on global level in the last 50 years. Areas where a particularly extensive warming has been recorded, are the Iberian Peninsula, Southeast Europe, northwestern Russia and the Baltic region. The rising of the mean annual temperature amounts to 1.4°C. The temperature increase has been more noticeable during the winter time than in summer. It is projected that by the end of the 21st century, the mean annual temperature increase will range from 2.1°C to 4.4°C and it is expected that it will be more pronounced in the summer period. In south Europe the temperature may rise locally by even 6°C¹².

A review of the climate change in Serbia so far as well as the climate projections for the 21st century has been given in the Serbia's First National Communication. Observed mean annual temperatures in the last 50 years show an upward trend in almost all of Serbia. An

increase of 0.04°C per year is evident, while in some areas in eastern and southeastern parts of the country a downward trend up to -0.05°C per year has been recorded. The highest increase in temperature has been observed in autumn season.

The rainfall amount observed in the period 1946-2006 has had an upward trend in most parts of the territory of Serbia. However, a decrease in rainfall amounts has been recorded in eastern and southeastern parts of Serbia. While an annual increase of maximum 16mm has been noted in the west, the amount of rainfall in the southwest of the country has dropped by 8mm per year.

Months	A1B1 2001-2030	A1B1 2071-2100	A2 2071- 2011
Dec/Jan/Feb	0.5-0.1	1.8-2.2	2.6-3.6
Mar/Apr/May	1-1.2	2.4-2.8	3.6-4
Jun/Jul/Aug	1.2-1.4	3.2-3.6	4.2-4.6
Sep/Okt/Nov	0.5-0.9	1.8-2.2	2.6-3.2
Year	0.8-1.1	2.4-2.8	3.4-3.8

Table 3: Temperature variations in degrees Celsius per season. Source: Ministry of Environment, Mining and Spatial Planning, 2010

Projections made according to the EBU-POM model predict that by the end of this century mean annual temperatures will increase in the range of 2.4°C - 2.8°C in the A1B1 scenario, respectively 3.4°C - 3.8°C in the A2 scenario. It is expected that the rising of temperature will follow a variable trend during different seasons (see table). In other words, a more significant temperature increase (3.2°C - 3.6°C) is expected in the summer months.

Months	A1B1 2001-2030	A1B1 2071-2100	A2 2071- 2011
Dec/Jan/Feb	-10 -5	-20-0	-15-5
Mar/Apr/May	15-15	-15-10	-30-0
Jun/Jul/Aug	-5-30	-30-5	-50-10
Sep/Okt/Nov	-10-20	-30-5	-30-10
Year	-5-10	-15-0	-15-5

Table 4: Precipitation variations in % by season. Source: Ministry of Environment, Mining and Spatial Planning, 2010

According to the same model, it is expected that by the end of this century the rainfall will indicate a downward trend. In the A1B1 scenario the precipitation will decrease

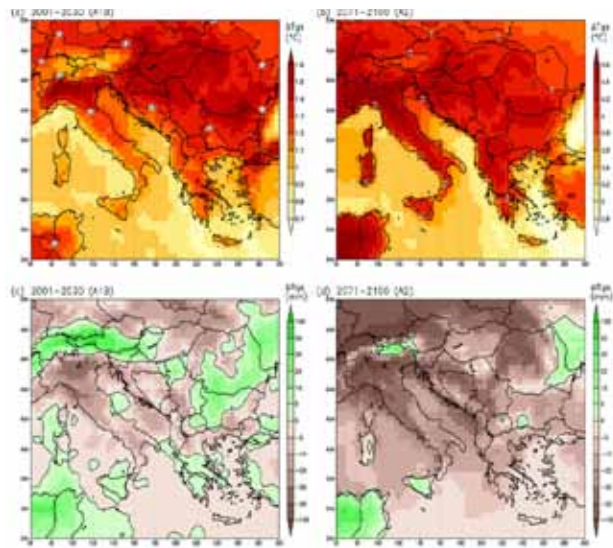
¹¹ See http://www.hidmet.gov.rs/latin/meteorologija/klimatologija_srbije.php

¹² IPCC, 2007

in the range of -15% to 0%, while in the A2 scenario it will drop by 15% to -5% per annum (see table).

According to the projections in the A1B1 scenario the temperature increase should be relatively consistent throughout all Serbia, although a slightly more pronounced upward trend is expected in the east along the Danube River and in the southwest of the country. In the same way, it is expected that the precipitation will follow a mild downward trend in the north of Vojvodina and in some regions in the southeast and east of the country.

In the A2 scenario, a quite consistent upward trend is also expected in regard to temperature variation in whole Serbia. Slightly smaller increase is predicted for the greater part of Vojvodina, some areas in west Serbia, in the Drina River Valley and in the southern regions. In this scenario the precipitation amount shows much more spatial variability. Namely, while in some regions the precipitation amount will increase by the end of the 21st century, it will drop in other parts of the country. A downward trend in regard to the precipitation will intensify from the northeast to the southwest, so that a decrease of precipitation



III.5. Mean annual temperature (above) and average annual precipitation (below) variation predictions for two periods (2001-2030 and 2071-2100) as per Ruml et al (2012)

is expected in the southwest of the country (-10 to -15%). Some more recent climate projections¹³ confirm similar general trends in temperature and precipitation variability, however with greater local disparities in case of climate change impact.

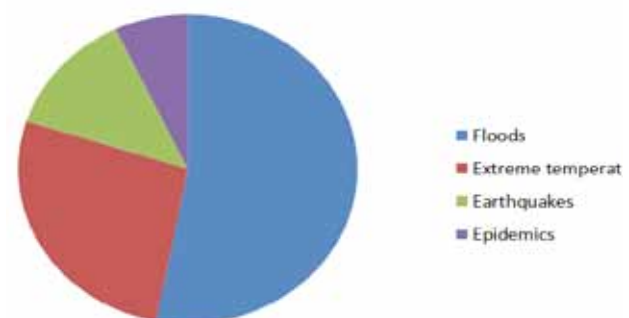
13 Ruml et al, 2012

2.3 NATURAL DISASTERS

2.3.1. The situation in Serbia and the SEE region

The entire region of southeast Europe has proven to be very vulnerable to natural disasters. Almost all countries in this region have been significantly affected throughout their history by the effects of floods, earthquakes, landslides, forest fires, droughts, heat waves, prolonged winter seasons, and hailstorms. The effects of natural disasters have been considerably intensified by the relatively poor economic situation in these countries. Also, the means and infrastructure necessary to adequately forecast and prevent disasters as well as to mitigate their impact have been noticeably degraded in the period of political instability in the Balkans.

Serbia has been greatly exposed to natural disasters, primarily floods. According to the International database on disasters (EM-DAT), in the period 2000-2011, Serbia faced a higher risk of natural disasters



III.6. Frequency of various natural disasters in Serbia in the period 2000-2011. Source: EM-DAT

than of technological accidents. Of all disasters, 62% were caused by nature. Among natural disasters, floods dominate with 55% of the total number.

	2004	2005	2006	2007	2008	2009	2010	2011
Privately owned forests	87	22	146	14360	152	957	223	1329
Public forest estate	115	30	348	7801	423	253	280	707
Total	202	52	494	22161	575	1210	503	2036

Table 6. Burnt forest areas in the period of 2004-2011 on the territory of Serbia. (Source: <http://webzrs.stat.gov.rs>, 2012.)

Floods affect a large number of people and cause great material damage, much more than all other types of natural disasters.

Disaster	Number of events	Number of death victims	Number of affected persons
Floods	8	4	51.290
Earthquakes	2	3	27.130
Extreme temperatures	4	5	70
Epidemics	1	-	170

Table 5. Overview of numbers of natural disasters, death victims and affected persons in period 2000-2011 on the territory of Serbia (Source: EM-DAT)

Notably, only natural disasters are mentioned above which in their scope and intensity have posed a threat to human safety and have caused great material damage. Additionally, extreme hydrometeorological events should not be ignored either. Even though they do not reach the critical scope of disasters, they can still have a significant impact on certain economic sectors.

Forest fires also commonly occur in Serbia and are a serious threat to people's safety as well as in terms of

causing great material damage. In the past decade, fires have covered significant areas of woodland. Particularly, the year 2007 stands out when over 22.000 ha of woodland were destroyed by fire (Table 6).

2.3.2. Trends in natural disasters

The number of natural disasters indicates a perceptible increase in Serbia. This number rose from an average of 100 disasters a decade in the early 19th century to 2,800 disasters a decade by the end of the 20th century (see Table below).

1900-1940	1960-1970	1980-1990	1990-2000
100	650	2000	2800

Table 7. Number of natural disasters by different decades. *average number of disasters per decade in the given period of time (National Strategy for Protection and Rescue in Emergency Situations).

The upward trend in natural disasters is especially apparent in occurrences that are caused by hydro-meteorological conditions. Thus, for instance, the number of disastrous floods has grown by the end of the 20th and at the beginning of the 21st century¹⁴. Significant flooding occurred in 1980, 1981, 1988, 1999, 2002, 2005 and 2006. The flood of 2006 exceeded the historical highest water levels on the Danube River.

¹⁴ Milanovic et al., 2010

2.4 NATURAL DISASTER VULNERABILITY

2.4.1. Current situation in regard to natural disaster vulnerability

Different economic sectors show various levels of vulnerability to natural disaster and extreme hydrometeorological events. One of the most vulnerable sectors is agriculture, followed by energy, water management and air traffic. The share of sectors

that depend on weather conditions in the gross national product of Serbia is considerably substantial and amounted to 47.18% in the year of 2005¹⁵. An overview of the loss estimates by sector caused by natural disasters is presented in Table 8:

¹⁵ World Bank

Sector/ Dangerous or extreme hydrometeorological events	Damage assessment by sector	
	Average annual economic loss in millions of dinars	Average annual loss in human lives
Agriculture/ floods	3.100 - 8.500	several to tens
Water management/ floods	approx. 1.960	-
Agriculture/ hail, heavy precipitation, strong winds	approx. 7316	several to tens from thunder stroke
Agriculture/ drought, frost	approx. 40.000	no losses
Energy production (heating energy)/ extremely low air temperatures	approx. 716	several to tens
Road maintenance/ snowfall, ice, freezing	approx. 3.500	-
Human losses on highways, regional and local roads caused by bad weather conditions vary annually from 105 to 131		
Commercial air transport	54 - 72	-
TOTAL	16.648 - 48.572	several to 160

Tabela 8. Procena gubitaka po sektorima koji nastaju usled nepovoljnih hidrometeoroloških događaja. Izvor: The World Bank, 2005.

The analysis shows that droughts also represent a major risk to many economic sectors. However, forest fires have neither been a subject of the World Bank study nor recorded in the International database on disasters. Forest fires represent a significant threat factor in Serbia and cause enormous damage. Aleksic and Jancic (2011) estimated that the total damage caused by fires in public forest estates in the period from 2000-2009 exceeded 36 billion dinars (Table 9).

2.4.2 Trends in natural disaster vulnerability

It is still difficult to talk about the dependence of such trend of natural disasters on climate change. Most experts agree that climate change will cause a greater occurrence of extreme climatic events, such as floods, landslides, and wildfires¹⁶. These hazards have also been recognized at the political level so that improvements of the prevention system, disaster defense and mitigation of their impact are regarded as priority by many national governments and international organizations. The concerns of the world community because of the

¹⁶ IPCC, 2012

Activity	RSD
Firefighting costs	44.498.395
Damage caused by fire	34.199.158.808
Troškovi sanacije, gajenja i zaštite šuma	2.211.105.203
TOTAL	36.454.762.406

Table 9. Damage caused by fires in public forest estates in the period of 2000-2009 (Aleksic & Jancic, 2011).

upward trend of disasters and the growing number of people killed have resulted in the adoption of the Hyogo Framework for Action 2005-2015. This action plan aims to build capacities in the fight against disasters. Serbia has launched, in accordance to recommendations of this plan from 2009, a comprehensive institutional and organizational reform of its defense system against disasters. This has resulted in adopting of a new Law on Emergency Situations¹⁷ and forming of a chief coordinating body, the Department of Emergency Management, and creating the National Strategy on Protection and Rescue in Emergency Situations (2011).

¹⁷ „Official Gazette of the Republic of Serbia“, issue 111/09

2.5 ADAPTIVE CAPACITY TO CLIMATE CHANGE IN SERBIA

The current adaptive capacity to climate change in Serbia is rather moderate. Development of an adaptation

system to new climatic trends has not yet been identified as priority in public policy. The political and

economic elite still does not recognize the seriousness of the situation and the vulnerability degree of the whole society in a volatile climatic situation. Several factors have contributed to such a state of affairs. Many years of political instability in the Balkans have led to the degradation of the state systems necessary to operate in such circumstances. Considerations of future trends and hazards that they carry for the society as a whole have been disregarded and have lost priority over actual political issues. The poor economic situation, additionally destabilized by the global economic crisis, has led to the impoverishment of professional and scientific capacity and resources that are of critical importance in such situations. It is not easy to tackle strategic planning in adaptation to climate change under such political and economic circumstances.

Yet, in the process of political transformation and by Serbia's gradual rapprochement with the European Union, its capacity for a strategic analysis of climate change issues has begun to improve. The current activities primarily focus on the processes necessary for a reduction of climate change causes (mitigation), while adaptation approaches are still neglected. Serbia has significantly improved its capacity in the field of

climate change monitoring, which primarily pertains to the Republic Hydrometeorological Service of Serbia that has become a respectable institution in the region and Europe with the help of the government and foreign donations. As such, it represents an important source of information for all sectors that might be affected by climate change. The Republic Hydrometeorological Service of Serbia regularly provides weather (Meteoalarm) and hydrological (Hydroalarm) forecasts, forest fire danger index (FWI) and UV index forecasts, announcements of hot and cold waves, and lots of other information. Unfortunately, it seems that this information is still unknown to competent decision-makers who would integrate them into their strategic documents. Surprisingly, many sectors vulnerable to climate change have still not recognized the need for better and more appropriate planning of adaptation measures. Taking into account the considerable potential of natural adaptation capacity of this region, Serbia could do much more to revise its sector policies in order to reduce the negative impact of climate change. Many things could be changed by using sustainable ways in managing natural resources and in adequately planning and managing the development of agricultural production.

3 Climate change impacts

Rising temperatures, spatial and temporal precipitation regime changes, as well as all other changes that entailed by these two climatic parameters could have a strong impact on many aspects of the society, economy and the general functioning of the state. The increasingly frequent adverse weather condition occurrences, as well as the possible increase of extreme natural phenomena with catastrophic effects, will certainly have a significant impact on all sectors and activities that depend on natural resources, their use and protection. Considering the fact that we can hardly imagine any activity that does not stand in some way in relation with the conditions in nature, we can comprehend the full seriousness of the

climate change impact. It is clear that in areas such as public health, agriculture, energetics, water resources management and environmental protection the climate change impact will be most directly and intensively felt. Therefore, these sectors are presented separately in terms of how they would be affected by climate change and what possible adaptation measures for their reduction would be. The selection of the above mentioned sectors has been carried out in cooperation with the "Climate Forum", a national network in Serbia, which is part of the "South East European Forum on Climate Change Adaptation".

3.1. WATER RESOURCES

3.1.1. Water resources in Serbia

The total runoff of internal waters on the territory of Serbia amounts to 500 m³/s, i.e. 16 billions m³ per annum. This is about 1500 m³ per capita which puts Serbia in line with poorer countries in relation to water resources. A tenfold greater amount of water (162.5 billions m³) comes to Serbia from the neighboring countries, en route our largest rivers Danube, Sava, Tisa, Drina, Nisava and Tamis¹⁸.

The current water consumption in Serbia is around 900 millions m³. The largest part of it is used for population supply (ca. 700 000 m³), while approximately 200 000 m³ is used for the industrial demands¹⁹. Water losses that occur during the water supply are estimated to be even 30% higher than in other European countries.

A substantial increase in water demand is expected in the near future. According to the estimates given in the

Water Management Basics (Institute "Jaroslav Cerni", 2001), Serbia would need from the year 2011 on around 5.6 billions m³ of water per year. A consumption of such an extent comes close to the borders of sustainability and represents definitely a challenge for the water resources management.

Additional problems in the use and conservation of water resources are caused by their distinct regional differences in Serbia in terms of size and spread of runoffs. Thus, areas with the highest population density and greatest water demand have the lowest internal runoff. Vojvodina, Belgrade and Morava Valley have a very poor annual runoff, while the mountainous parts of Serbia have much more water resources, especially in the southwestern and western parts of the country²⁰.

Water quality in Serbia is relatively unfavorable, which is primarily caused by the absence of water treatment systems so that municipal and industrial waters are mainly discharged without being treated before in

¹⁸ Water Management Basics of the Republic of Serbia, „Official Gazette of the Republic of Serbia“, No. 11/2002

¹⁹ Source of information: Statistical Office of the Republic of Serbia: <http://webzrzs.stat.gov.rs>

²⁰ Isailov & Srna, 2001

any way. The level of wastewater treatment facilities is currently very small in Serbia (only slightly more than 10% of the population has systems for wastewater treatment), whereby some systems are not functional or work with inadequate capacity.

Generally, the worst water quality is to be found in the channel systems in Vojvodina, whereas the Lim and Drina Rivers have the best water quality²¹.

3.1.2. Climate change impacts on water resources

The problem of possible climate characteristics in Serbia has not been explicitly mentioned in the existing strategic documents of the Serbian water sector, above all, in the Water Management Basics of the Republic of Serbia. However, great attention has been devoted to this issue, so that in 2000 already, the study "Water for the 21st Century" (*Jaroslav Cerni* Institute for Development of Water Resources, 2000) was elaborated on behalf of the Ministry of Agriculture, Trade, Forestry and Water Management. The main purpose of this study was to examine the Republic's possibilities as to how to respond to the growing pressures in relation to this natural resource, i.e. to predict certain "limits" (for the next 100 years) of country's water demands, further to check on available water resources and ways of their exploitation and to provide specific guidelines for a rational "closure" of the water balance, all under the circumstances of possible climate change effects. In addition to this, in 2010 and 2011, the Ministry of Agriculture, Trade, Forestry and Water Management - Republic Directorate for Waters has financed the feasibility study "Impacts of Climate Change on River Regimes in Serbia".

The 2011 research project „Assessment of Climate Change Impact on Water Resources of Serbia“ was financed by the Ministry of Education and Science and project participants included the Ministry of Agriculture, Trade, Forestry and Water Management - Republic Directorate for Waters and the public company Electric Power Industry of Serbia. Within the project framework, the impacts of climate change on water resources were analyzed in different climatic scenarios thoroughly discussed within the interdisciplinary project "Research of Climate Change and Its Impact on Environment: Impact Monitoring, Adaptation and Mitigation", funded

solely by the Ministry of Education and Science. The scientific contribution of the project should be to the modelling of the climate change effects on water resources and its practical implementation.

Apart from the national projects, the climate change effects on Serbia's waters have been analyzed within various international projects, also with participation of researchers from Serbia. One of them is „CCWaterS – Climate Change and Its Impact on Water Management“²². Eighteen institutions from Austria, Bulgaria, Greece, Croatia, Hungary, Italy, Romania, Serbia and Slovenia participate in the project. The Project CCWaterS will identify and assess the climate change impacts on water availability and water supply security.

The most comprehensive analysis of potential climate change in relation to waters of the Danube River basin was done within the project "The study of Danube River – Adaptation to Climate Change" on behalf of the International Commission for the Protection of the Danube River" (ICPDR), carried out by the Ludwig-Maximilian-University in Munich in 2011 (Mausser et al, 2012). Numerous international and local projects and studies (over 60) on climate change in Central and Southeast Europe, the Danube River Basin and its tributaries area were analyzed within this study. The study aimed to elucidate the current state of knowledge in regard to the climate change in this part of Europe and to suggest possible adaptation measures on potential climate change that would be harmonized by all countries in the Danube River Basin and serve as a basis for future cycles within the suitability framework for the Water Management Plan of the Danube River Basin.

The European directives are being transposed to the national legislation during Serbia's accession period to the European Union.

The EU member states deal with climate change issues in a qualitatively good way when it comes to river basin management as stipulated in the first stage of the River Basin Management (RBM) under the EU Water Framework Directive (WFD). In the first cycle of the River Basin Management, the Water Management Policy Paper has put a special emphasis on ensuring that the set programme of measures are sufficiently adaptive to future climate conditions (so called

²¹ Agency for Environmental Protection, 2011

²² <http://www.ccwaters.eu/>

“climate-check” of the programme of measures that is being carried out based on the knowledge base, information and data available).

In the second and third RBM stage, the climate change effects should already be fully integrated in the management process of river basins. The adaptation approach in managing river basins within the EU Water Framework Directive involves 1) efficient long-term monitoring (to identify climate change indicators in order to respond to them in a timely manner), 2) assessment of a potential additional impact of climate change on the already existing anthropogenic pressure, and 3) the inclusion of this information in the process of setting up (particularly for suggested measures which will have a long-lasting impact). The EU member states are thus at the very least expected to demonstrate that climate change projections have been considered in the assessment process of pressures and impacts, in the monitoring programs, and in the choice of measures.

In the Floods Directive of the European Union, climate change should be considered already in the first stage within the preliminary flood risk assessment, based on information available, but also in the subsequent planning stages to follow, during the revision and updating of the preliminary flood risk assessment and flood risk management plans.

The previous analyses of the water for various use, had low reliability of trend predictions. Generally, however, most of the models had predicted a reduction of water resources in this region as result of an increase of average temperatures and decrease of precipitation quantity. This is especially related to the summer months, when a substantial reduction of available water resources could be expected.

A higher water demand in the whole Danube River Basin could have significant consequences of available water resources in Serbia, bearing in mind that of all available water resources in the territory of Serbia over 90% comes from the upriver territories.

In the field of water defense, there are no explicit trends, but a substantial number of studies predict an increase in intensity and frequency of flooding, particularly in the winter season. On the other side, there are studies that do not foresee such a trend, so that a low level of reliability must also be taken into account

when it comes to projection of adaptive measures. In addition, certain studies indicate a potential increase in frequency of torrent floods in smaller basins, but the reliability of these predictions is also quite low, primarily because of the shift from global models to regional ones and the accuracy of this process when smaller basins are concerned.

The projected increase of water demands due to climate change, as well as decrease in precipitation amounts resulting in lower runoffs in rivers, will lead, especially in the summer, in low flow periods, to a higher pressure on surface waters in regard to their quality. These pressures are most significant for water bodies which are then, because of natural characteristics, extremely poor in water in the low flow periods, mainly in the summer. This is mainly valid for rivers in the south of Serbia (South Morava Basin), but also rivers in some other parts of Serbia (Central Serbia – Sumadija, eastern Serbia, etc.).

3.1.3. Adaptation to climate change

The basic adaptation measures refer to the development of basic planning documents in the domain of waters. These measures ensure a compliance of all other measures in the individual branches of the water sector.

The Law on Waters of the Republic of Serbia („Official Gazette of the Republic of Serbia”, 30/10) foresees the development of following strategic and planning documents to govern the water sector:

- Water management strategy
The water management strategy is a planning document that sets a long-term direction of water management and contains: an assessment of the current situation in water management, objectives and guidelines for the water management, implementation measures to achieve the set objectives and a projection on the development of water management. This document has to be harmonized with other strategic documents on the Republic's level in the field of spatial planning, sustainable development, utilization of natural heritage and resources, and environmental protection. The strategy should be adopted during the year 2012.
- Water management plans
Plans for water management are developed for a period of six years and should be adopted, in compliance with the above strategy, for the Danube River Basin and for other water regions

(Sava River, Belgrade, Morava River, Danube River, Srem, Backa and Banat, Kosovo and Metohija). The water management plans contain, among other things, in most general terms, the analysis of current state of water resources, environmental objectives for surface and ground water, as well as a presentation of the approved program of activities, measures as well as ways and means as to how the set goals will be achieved in all areas of water management (water usage, water protection, and protection from harmful effects of water).

In all these planning documents, but primarily in the Water management strategy, which should foresee the situation until the year 2030, it is essential to define possible solutions for all areas of the water sector, with the special attention paid also to measures that would ensure adaptation to potential climate change effects.

Specific adaptation measures are given for three main areas, for the water usage, for protection of harmful effects of water and for the water protection.

Water usage

Increase of retention capacity

As an increase of water usages is expected, the relatively unfavorable spatial and temporal allocation of water resources on the territory of Serbia has to be improved. This can be primarily done by building of reservoirs, in which water could be retained in periods of high water and used later on whenever it is needed and for various purposes: water supply for population demands, irrigation, industrial needs, improvement of low flow waters, for hydro-energetic facilities and similar. Serbia has already built over 30 large dams with water accumulations greater than 10 million m³, and a larger number of smaller dams and accumulations. Total accumulation capacity is more than 6 billion m³.

The country's existing planning documents, above all, the Water management basics of the Republic of Serbia have foreseen the construction of a significantly large number of new water accumulation objects, however, due to the poor economic situation, only a small number of these facilities has been built in the meantime (Prvonek) or are currently in the finishing phase of construction (Stuborovni, Selova).

According to the new insights about the water needs up to the year 2030, bearing also in mind the possible climate change effects, the Water management strategy has to define a certain number of new accumulation facilities, which would improve the total retention capacity in Serbia and allow a better spatial and temporal allocation of waters and ensure more flexibility and adaptability to future changes.

Water accumulation, apart from the surface accumulation, can also be gained by infusing water into underground aquifers by different methods and for different usage purposes. This would be possible in Vojvodina (accumulation of underground aquifers that are now threatened by over-exploitation) or in river valleys (Great Morava, etc.) by building multi-purpose systems.

Transferring water from water-abundant regions into water-deficient areas

Some areas in Serbia are abundant with waters comparing to their needs (above all in the Drina River Basin and its tributaries: Uvac, Lim, etc.), while in some other areas there are already apparent substantial problems with sufficient supply of quality water (Sumadija and Central Serbia, South Serbia, etc.) This requires consideration of possibilities of transferring certain quantities of water from water-abundant to water-deficient areas. This can first and foremost be done by transferring water from the Uvac River and, probably, Lim River to the Morava River Basin by accumulating in the Rzav River Basin.

In addition to this, as already mentioned, it is necessary to solve the problem of water deficiency in the central parts of Vojvodina by transferring water from its borders, such as: In Backa from the Bezdan-Apatin region or in Banat from Kovin-Dubovac area. There are similar problems in other parts of Serbia: Sumadija and Morava Valley, far South Serbia, so that it is necessary to find adequate solutions for these regions, too. Naturally, all these regional systems, by which a long-distance water transfers might be needed require substantial financial means and detailed and long-term planning.

Measures to reduce specific water use in industry, irrigation and the like

Due to the very difficult current economic situation, the water consumption is very limited for the purposes of industry, irrigation in agriculture, etc. Much larger water

quantities were used in the past for such purposes. The planned increase of production in the process of the domestic industry recovery, as well as areas of land under irrigation, inevitably leads to substantial increase of water demand. Therefore, in the planning process of new industrial and irrigation systems it is necessary to pay attention to implementation of technologies with the least possible consumption of specific waters. This will be especially important in the case of any significant reduction of water availability, particularly during the low flow periods (summer months), when the needs of irrigation water at the highest level. Besides, it should be noted that irrigation is the most significant “irreversible” water consumer, i.e. in the process of irrigating over 80% of water balance are used for this purpose.

Loss reduction in the water supply systems

Currently, the water losses in the supply systems are considerable in a large number of towns amounting to 20-30%, and even 50% in some towns. The reasons for that lie in poor maintenance of water supply networks and installations, above all because of the low price of water and the economically weak population. To cover the losses, it is necessary to draw considerable amounts of high quality water, which, apart from being uneconomical, puts additional pressure on already endangered water resources in certain areas. It is, therefore, necessary to enforce measures that would resolve this issue. This could be done, in the first place, by the gradual price increase of water and communal services to the economically justified level, making sure that all necessary measures of social protection for the economically most vulnerable consumer categories.

Improving monitoring and other non-investment measures to combat droughts

So far, the fight against water deficiency in Serbia, contrary to the quite regulated defense system against high waters, has been based mainly on mitigating its consequences, i.e. helping the agriculturist to alleviate the effect of droughts. Insufficient attention has been paid to improvement of the projection and monitoring system of this phenomena, preparation of plans for mitigating effects of drought and distribution of available water recourses. Also, there is a need to improve the cooperation with other countries with the goal to exchange experience in combating droughts.

Defense against harmful effects of water

The strategy of protection from harmful effects of water has to be founded on following principles in the time to come:

- coordination – harmonized activities of responsible organizations at the local, regional and national level;
- integral approach – ensuring protection of water by combining financial and non-investment measures;
- prevention – preventing construction in areas susceptible to flooding, mainly by producing maps of blue zones and installing measures that discourage building in those areas, which comply with the principles given in the EU Flood Directives as well as citizens’ education about floods and existing defense limitations;
- reality – awareness that there is no absolute defense against flooding, and that floods fall into the category of stochastic processes.

Measures that should be considered in this field:

1. Conception of defense plans against floods of international waters which are conform at the level of large river basins (Danube, Sava, Tisa Rivers, etc.)
2. Increase of the safety level of flood defense, especially for valuable areas (biggest agglomerations: Belgrade, Novi Sad, etc.), largest industrial centers, largest thermo-energetic facilities, etc.)
3. Extension of retention areas in river basins by contracting cassettes, accumulations, etc.

As stipulated in the Law on Waters, all these measures must be integrated in the Plan of flood risk management. The Law on Waters envisages that this plan will be adopted until the year 2017. In this plan, all measures for the most important flooding areas in Serbia will be defined that have been identified in the Preliminary assessment of flood risk in 2011. All plans will be based on vulnerability and flood risk maps, which are currently being elaborated.

Water protection

The basic measure that regulates the issues of water protection is the construction of wastewater treatment plants, for settlements as well as for industry. In the water management plans, the construction dynamics of wastewater treatment plan will be given, so that

all settlements with more than 2000 inhabitants will be included into wastewater treatment. Besides, as an additional measure of water quality improvement, particularly in the period of low flow, the plan foresees deliberate discharge of upriver accumulations in order

to improve the quality of water in the valleys, and eventually, in further time intervals, the transfer of water from water-abundant basins (Drina River Basin) to water-deficient basins (Morava River Basin).

3.2. PUBLIC HEALTH

3.2.1. Health system of Serbia

The ongoing overall changes in the social order since 2000 have also affected the health sector. The old system of social protection inherited from the socialist system has proved inadequate in present conditions. Such a health system was based on state subsidies, but the country in the stage of transformation was no longer able to provide substantial resources for its functioning. For this reason, the overall situation in the health sector suddenly deteriorated, which also affected the quality of health services. Since 2000, extensive reforms have been initiated as well as the restoration of the health system. The reforms were based on rationalization, decentralization and modernization of the system's organization as well as on introduction of an interdisciplinary and intersectoral approach in solving problems in health care.

At the top of the health care system in Serbia is the Ministry of Health that defines the health policy, sets standards of health protection and ensures quality control. Next to the Ministry at the top of the health system pyramid is the National Public Health Institute of Serbia „Dr Milan Jovanovic Batut“ and the Republic Health Insurance Fund. The National Public Health Institute monitors the state of public health and the overall situation in the health sector; based on this, it provides professional support in defining and managing the health policy. The Republic Health Insurance Fund finances the health protection activities at all levels.

The national health system consists of 301 health institutions²³. In Serbia, there are 116 health centers, 16 general hospitals, 22 medical centers, 37 specialized hospitals, 23 public health institutes, 4 clinics and 4 clinical centers.

Expenditure on health care was around 7% of the GDP in the last decade²⁴. According to this parameter, Serbia ranks shoulder to shoulder with the West European countries. However, due to the low GDP, the actual expenditure expressed in money should be taken into account in order to gain a more realistic picture. Thus, according to average annual expenses for health care, with 336\$²⁵, Serbia falls into a group of European countries with the lowest spending on health care. The number of health workers per capita in Serbia is below average in Europe - 272 doctors and 528 nurses per 100,000 inhabitants²⁶.

3.2.2. State of public health

The global improvement of the health protection in the 20th century positively influenced the state of public health in Serbia. Contagious diseases, which represented a major problem in the past, are nowadays fully under control. In addition to this, the health of mothers and children has been significantly improved and infant mortality reduced. Accordingly, the water and food supply of adequate quality had a substantial influence on the overall health condition²⁷. However, the general state of health has deteriorated over the last decade. The reason for this is primarily lifestyle changes that have been incited by risk factors, such as unhealthy diet, smoking, alcohol and psychoactive substances abuse, obesity, poor physical activity, stress, etc. Due to all of this, chronic, non-contagious illnesses currently represent the greatest problem in regard to public health of Serbia. Most of them, which make over two thirds of all health problems, are cardio-vascular illnesses and malignant tumors²⁸.

²⁴ Ibid.

²⁵ Ibid.

²⁶ Ibid.

²⁷ Public Health Strategy of the Republic of Serbia

²⁸ National Public Health Institute of Serbia, 2011

²³ National Public Health Institute of Serbia, 2008

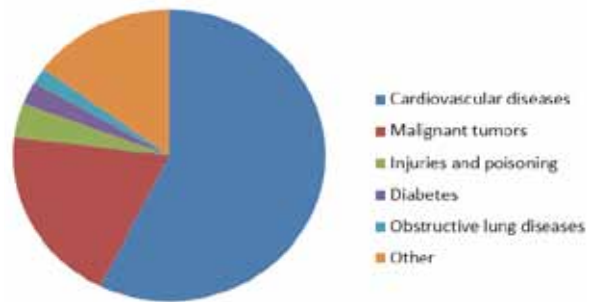
As per data from 2010, life expectancy in Serbia is 74 years. According to this parameter, Serbia is close to the Eastern European countries, but considerably behind the Western European countries. It is important to mention that life expectancy has followed an upward trend in Serbia in the last decade. A similar situation can be found with the other essential public health parameter, the infant mortality. The 1997 infant mortality rate in Serbia was 12.1 infants of 1000 live births, while in 2007 it dropped to 7.1/1000. The rate was significantly higher than in 27 EU member states, where the infant mortality rate was 4.5/1000, although some Eastern European countries (Bulgaria, Romania) had higher infant mortality rates²⁹.

In the period from 2001 to 2010, the overall mortality rate in Serbia rose by 6.1%. The most evident increase in the mortality rate was recorded with patients suffering from diabetes (40.8%), obstructive lung diseases (23.6%) and malignant tumors (21.4%). On the other hand, in case of injuries and poisoning, substantial reduction (-23%) of the mortality rate was observed over the same period of time.

The cardio-vascular illnesses, which make 54.7% of all causes of death, are the leading cause of death in the Republic of Serbia³⁰. From 2001 to 2011, the mortality rate of patients suffering from cardio-vascular illnesses increased by 9.2% for women and 2.7% for men. In 2000, 21.1% of the total burden of ischemic heart diseases was related to hypertension. The number of persons that take antihypertensive drugs increased from 46.5% in 2000 to 51.3% in 2006. In this period of time, the number of persons who began taking drugs against hypertension was increased by nearly 18%³¹.

During the first decade of the 21st century, around 23,730 persons contracted malignant diseases in Serbia each year, with the growing trend. The incidence rate was higher in males with these diseases and amounted to 282.8/100,000, while in females, it amounted to 240.5/100,000³².

Over 90% of skin cancers emerge on parts of body regularly exposed to sunlight or ultraviolet radiation. Skin cancer is directly attributable to UV radiation. There



III.7. Most common causes of death in the Serbian population. Source of information: National Public Health Institute of Serbia (2011)

are two types of skin cancer: basal cell carcinoma and squamous cell carcinoma. The basal cell carcinoma develops from the deepest layers of the epidermis and represents the most common form of skin cancer. It makes 75% of all skin cancers. The main characteristic of this type is its slow growth. Although basal cell carcinoma rarely metastasizes, it grows locally with invasion and destruction of local tissues (muscles, bones, tendons). More than 90% of such tumors grow in the areas exposed to sunlight, mostly on the face and neck. It occurs more frequently in elderly people due to longer exposure to the sun. Squamous cell cancer is more aggressive type that metastasizes much more often. It develops in the main part of the epidermis and it develops also on skin that has not been exposed to the sun. Due to their occupation, outdoor workers such as farmers and construction workers are the most prone part of the population to skin cancer. The annual increase of melanoma incidence in our country, according to the data from the National Public Health Institute „Dr Milan Jovanovic Batut“, was 5.3% in 2004 and 15% in 2005, which was much higher than the world standards, where the annual increase was between 3 and 7 percent. Melanoma can occur at any age, but the highest prevalence is for persons from 45 and 55 years of age, equally for men and women. Although the exact cause has not yet been discovered, the main risk factors seem to be extensive sun exposure (especially occasionally, i.e. “recreational” exposure to sun light)³³. The prevalence of this, as well as other skin cancers has steadily grown, due to the thinner ozone layer.

More than half a million people in Serbia suffers from chronic, non-contagious respiratory diseases. The mortality rate from chronic obstructive pulmonary diseases (COPD) increased by 11.8% during the last

29 Source: <http://epp.eurostat.ec.europa.eu>

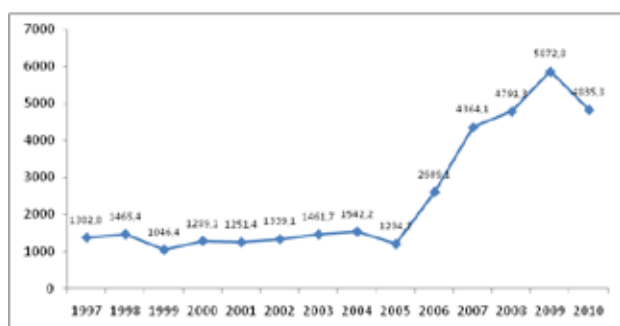
30 National Public Health Institute in Serbia, 2011

31 National Public Health Institute of Serbia, 2008

32 National Public Health Institute of Serbia, 2011

33 Source: Institute of Oncology and Radiology <http://www.ncrc.ac.rs/strane/rakkoze.htm> (Retrieved on 23 Feb. 2012).

ten years, from 17 per 100,000 inhabitants in 1997 to 19 per 100,000 inhabitants in 2007. The most important risk factors are active smoking, genetic predisposition, allergy, viral and fungal infections, and environmental factors including climate. Different types of allergies represent a great threat to public health and the quality of life. The global upward trend of chronic allergy incidence is present in Serbia. Unfortunately, there is no official statistics on the number of people who suffer from allergies, so that no reliable statistical data based on which any assertions could be made in regard to the prevalence or trend of these diseases among the population. Nevertheless, most local experts agree that there has been a substantial incidence of allergies. It is estimated that 20-30% of inhabitants suffer from some type of allergy.

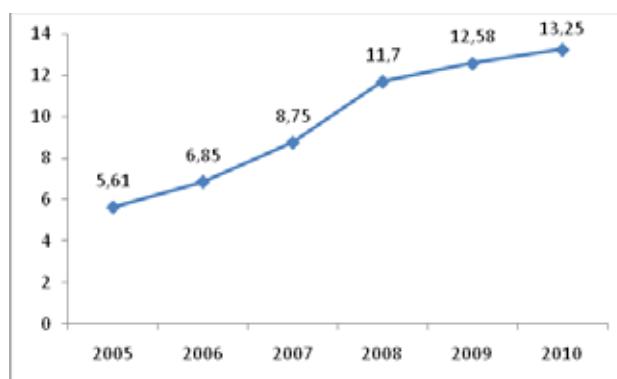


III.8. Variation of incidence (number of incidents per 100 000 inhabitants) of contagious diseases in Serbia in the period from 1997 to 2010. Source of information: National Public Health Institute of Serbia.

Contagious diseases still represent a dangerous threat, although a great progress has been made in the last decades in regard of prevention. The incidence variations of contagious diseases has shown on an annual level a significant increase in the last five years, but this trend cannot be regarded as a real epidemic situation. Namely, higher incidence rates of contagious diseases in the first decade of the 21st century are primarily results of improved reporting and recording of contagious diseases.

Respiratory tract diseases, primarily pharyngitis, tonsillitis and influenza dominate among contagious diseases. Of infectious diseases, the leading causes of death in Serbia in the past decade included HIV, tuberculosis, sepsis, and bacterial meningitis. It should be mentioned that in the last several years the incidence of tuberculosis has substantially decreased, so that nowadays it is below 20 incidents per 100,000 inhabitants.

The highest incidence rate of animal-borne diseases is found with Lyme disease. In 2010, there was an incidence of 13.25 cases per 100,000 inhabitants. The incidence of this disease has been growing over the past several years, but it is difficult to say whether this is caused by better reporting means or by actual prevalence of the disease. Apart from Lyme disease, malaria, viral encephalitis, leishmaniasis and hemorrhagic fever are rare in Serbia. Generally, only some isolated cases of these diseases are diagnosed annually, while malaria incidence rate is slightly higher. On average, there were 10 cases per year in the last decade. It should be mentioned, though, that all cases were tied to travel abroad.



III.9. The incidence of Lyme disease in Serbia from 2005 to 2010. Source of information: National Public Health Institute of Serbia, (2011).

The outbreaks are usually alimentary epidemics or outbreaks of diseases transmitted by food, and they have made 65% of the overall number of epidemics. Outbreaks of diseases transmitted by water accounted only for about 1.5% of the total number of epidemics in the past in Serbia³⁴.

The availability and quality of water for drinking and cooking, as well as the quality of air are important factors that are affecting public health. The water supply situation in Serbia is relatively good and an extremely high percentage of the population (over 95%) has a water connection in their house or apartment. The water quality is generally following an upward trend, i.e. the number of unsatisfactory samples has reduced. However, there are still problems with the water quality in certain regions of the country, especially in the Pannonian Plain. Also, here is neither an adequate control system for the whole territory of Serbia, nor are all parameters being monitored that could influence the

³⁴ National Public Health Institute of Serbia, 2008

public health. The quality control of food products has been substantially improved in the last couple of years and there have been less bacteriologically inappropriate samples found. The air pollution shows major differences on specific locations in Serbia. Lately, there has been a significant increase of spots where the quality of air is being measured, but their number is still insufficient.

3.2.3. Climate change impacts on public health

It is important to remember that the health is not only a physical category; it is also a social and mental wellbeing. With such an understanding of health, it is clear that effects of climate change can be complex and far-reaching. The impact could be direct, when changed weather conditions cause health problems, or indirect when climate change affect some processes and resources which are essential to human health, such as quality and availability of water, quality of air, food production, transport, or infrastructure.

For many years now, climate change impact on health has been attracting attention of experts and international public in general. Although, there are many unknown factors and concerns, the experts have reached a certain level of agreement about general characteristics of these impacts. According to Campbell-Lendrum et al (2007) these general characteristics are:

- health risks are different, global and probably irreversible in terms of temporal scale of humanity;
- potential impact on health, and consequently on humanity as a whole are enormous;
- risks are inequitably allocated and more concentrated in the development countries;
- many projected impacts could be avoided or mitigated by various interventions.

Most experts agree that climate change will not cause new types of diseases, but it will rather affect the factors that define a phenomena and incidence of already existing health³⁵.

Public health is generally very dependable on weather conditions, but diseases that are expected to be mostly influenced by climate change impacts are³⁶:

- asthma, respiratory allergies and diseases;
- malignant diseases;

- vascular diseases;
- nutrition and diseases caused by nutrition;
- diseases and mortality caused by heat stress;
- mental health disorders, disorders caused by stress;
- neurological disorders;
- vector and zootomic diseases;
- diseases transmitted by water;
- diseases and mortality caused by weather conditions.

Increased prevalence of asthma and other diseases of the respiratory system is expected due to a greater exposure to pollen (prolonged vegetation season), to air pollution and increased concentration of dust due to droughts. The increased intensity and duration of UV radiation will directly affect an increased incidence of malignant diseases, above all skin cancer. Also, there is a possibility that climate change will have an impact on an increase in human exposure to carcinogenic materials and thus, indirectly, will cause a higher incidence of these diseases. A more frequent occurrence of extreme weather conditions will probably cause a higher morbidity rate in cases of cardiovascular diseases, but also on epidemic infections that are related to these diseases.

Stress situations that might occur as effects of the climate change (extreme weather conditions, natural disasters) will certainly affect the mental health of people and a general stress increase. Irregular climate conditions may induce a change of spread and incidence of some infective diseases, i.e. vectors that transmit such diseases. The impact on food and water and diseases related to them will not only reflect in the reduction of their availability, but also in their lower quality and higher contamination level. Finally, extreme climatic events could have disastrous effects and cause enormous casualties in the population. Bearing all these potential impacts in mind, public health seems to be very exposed to climate change and its effects could cause serious problems in the functioning of the society as whole. The climate change impacts on other sectors will also, directly or indirectly, affect the public health. Increasing poverty reduced economic potentials in some areas or for certain groups of people, as well as lower availability to specific services may quite adversely affect the social and economic population status, and thus, their health.

³⁵ Sumet, 2009

³⁶ Portier et al, 2010

A detailed analysis and prediction of potential climate change impacts on health has never been made. Also, none of the existing strategic documents in the field of health is considering risks that are caused by climate change. Although, the draft of the document "Health policy guideline for strengthening the health care system in Serbia from 2010 to 2015"³⁷ recognizes climate change as one of the factors that determine health, but it does neither estimate the expected effects on it nor suggests any adaptation measures. For the time being, there is only some research on the effects of meteorological phenomena on human health, more precisely – on specific health problems. Thus a large dependency of stroke incidence and weather conditions has been observed on a sample of patients admitted to the Clinical Center in Kragujevac³⁸. According to the analysis of a sample of 4700 patients the highest incidence of stroke occurs during the biometeorological phase of warm and dry weather in cyclone areas and during a phase of wet and cold weather in anticyclone areas. The same team of authors has elaborated also a biometeorological impact analysis on the mortality from cardiovascular diseases, also on a sample from Kragujevac³⁹. On a sample of 631 death cases caused by cardiovascular diseases, a significant correlation has been observed between mortality caused by cardiovascular diseases and specific biometeorological phases. According to this study, the largest number of deaths (30%) has occurred in the biometeorological phase that is characterized by a strong vertical development of clouds with rainfall, thunder and strong winds, sudden drop in temperature and pressure rise. In any case, further research is necessary in order to determine more precisely the relation between weather conditions, i.e. biometeorological phases and incidences of some chronic, non-contagious diseases. It is obvious that some correlations exist and that by certain prevention measures and meteoprophylaxis the mortality could be reduced and the general public health improved.

It is generally known that weather conditions have a great influence on the mental condition of people and on their mental health. On a sample of the town of Kragujevac, the correlation has been shown between

suicide and biometeorological phases⁴⁰. Most suicides have occurred in the phases of stable sunny weather and sudden impact of weather fronts.

On a global level, great changes in the dynamics and spread of contagious diseases are expected due to climate change. Incidences of many diseases have been observed in Serbia that may change their epidemic characteristics because of the climate change effect. These are, for instance, malaria, dengue fever, West Nile fever, chikungunya fever, Lime disease, and leishmaniasis. There is still no reliable evidence that any of these diseases is expanding in Serbia due to climate change, but given the global trends it is to be expected. Also, due to the expected, more frequent floods and other natural disasters a more frequent and extensive occurrence of water and food borne epidemics.

These are just some examples of direct impacts of weather conditions on human health. The question is how the expected climate change will affect human health and it is not possible to generalize this impact considering the specifics of some diseases. It is somewhat more certain that climate change will have substantial indirect effects on human health. This is primarily related to deteriorating socio-economic conditions in some regions, availability to wholesome water and healthy food, as well as access to health care services.

The recent events in early 2012, when Serbia and the neighboring countries have been struck by extremely cold weather, show best the severity of indirect climate change impacts on human health. Extremely cold weather followed by heavy snow in January and February of 2012 took 23 lives. During this period of time, a large number of people were cut off from main roads, so that their basic food supply was threatened. In addition, they were cut off from the access to the basic health care services, which caused the highest number of victims (impossibility of timely interventions of health care institutions). The population's access to health care institutions in Serbia is anyway very low. Every seventh household is more than 4 km far away from the nearest outpatient center, and every third from the health center. The worst access to health care institutions has the poor population, so that more than a third of them live more than 4 km away from the nearest outpatient

³⁷ <http://www.minzdravlja.info/downloads/Zakoni/Strategije/Nacrti/Nacrt%20Strategije%20Primarne%20Zdravstvene%20Zastite.pdf>

³⁸ Milojevic et al, 2011

³⁹ Gajic et al, 2011

⁴⁰ Gajic et al, 2011

center. In the case of the richer part of population this number is only 3.4%. Only 32.9% of poor people have their own physician, while in the richest segment of the populations this number amounts to even 62%. The access possibility to health care services varies considerably also by the regions in Serbia. The rural areas have on average a lower access to health care services; furthermore the largest percentage of poor population lives in these areas. Considering all above, it can be predicted that the health condition of the poor population in the rural, remote areas will be in danger by the expected climate change effects. In some parts of Serbia, extreme weather conditions and natural adverse phenomena (heat waves, droughts, floods, and cold waves) can significantly deteriorate the hygienic conditions, reduce accessibility to food and drinking water, as well as to health care services. All this, of course, leads to impairment of the state of public health.

3.2.4. Adaptation to climate change in the field of public health

Since strategic documents in this field do not sufficiently consider the problem of climate change and its potential impact on public health, the process of adaptation should start from the preparation of strategies and adaption plans, i.e. from the revision of existing strategic documents.

Adaptation at a strategic level

The vulnerability of public health to climate change in Serbia, especially in certain areas and among some population groups, is high. This requires a serious and systemic approach in dealing with this problem. Generally, the capacity of the national health system is sufficient to coordinate such a process and to carry it out in an adequate manner. The Ministry of Health, as the highest institution in the health system, should be in leading position and coordinating the national adaptation plans. Naturally, the process should also involve professional institutions and organizations, primarily public health institutes, medical institutions and scientific-educational institutions in the field of medicine, Red Cross Serbia and other relevant health institutions. As public health greatly depends on economical and social factors, the whole adaptation planning process should also include other institutions whose activities involve segments that may be of importance for health, such as water supply, food production, infrastructure (roads, energy infrastructure),

and actions in emergency situations. The Republic Hydrometeorological Service of Serbia is a very important institution in this process, since, as part of its activities, it already gives biometeorological predictions, i.e. projections of meteorological conditions from the aspect of their impact to public health.

Using strategic papers related to adaptation plans in the field of health, it is necessary to:

- develop an organizational structure for the process of adaptation to climate change in the field of public health that would include the assignment of responsible institutions and forms of their cooperation;
- define possible impacts of climate change in the field of health care;
- define health problems where the most significant impact of climate change is to be expected;
- define priorities for further research of the climate change impact on health;
- define groups in the population whose health may be mostly affected in the process of climate change (preliminary it could be said that these groups are the rural population living far from larger urban centers, older population, poverty-stricken population and outdoor workers);
- define the most vulnerable areas in Serbia (distant rural areas, regions prone to flooding and torrents);
- develop a set of measures for mitigation of certain problems;
- develop a system of notification and warning for population with health problems;
- integrate and harmonize adaptation plans in the field of health care with the national plan/strategy of adaptation to climate change
- consider development of immediate action plans for all types of disasters

Development of professional and scientific capacity to monitor climate change impact on health

Although the awareness of the need to keep abreast of health statistics has been substantially improved in Serbia, primarily because of the National Public Health Institute of Serbia “Dr Milan Jovanovic Batut”, appropriate statistics that would follow the relation of meteorological and climatic conditions and health parameters still do not exist. The development

of databases which would include information integrated from public health institutes, emergency and hydrometeorological services would represent a very important resource for monitoring reactions of public health on climate change, and thus an important foundation for adaptation planning.

Development of intersectoral cooperation

As it has been repeatedly pointed out, public health depends on several factors. Therefore, it is necessary to include representatives from several sectors in the adaptation process in the field of health care and to ensure adequate communication and coordination among them. For the health care, significant other sectors are agriculture, water management, infrastructure, and energy. .

Development of regional and international cooperation related to adaptation to climate change

The climate change will most probably have similar characteristics as in the neighboring countries. Therefore, it is useful to establish a network and exchange experiences in finding solutions to the many problems of public health. The need to cooperate internationally is especially pronounced in the field of contagious diseases. Bearing in mind that it is expected that climate change will affect the spread of some already existing contagious diseases as well as incidence of some new one, it is necessary to continue with the development of international cooperation in this field.

Development of public awareness about climate change impact on health

The level of awareness is still very low in Serbia, among policy and decision makers, as well as among the population. For a successful implementation of adaptation measures, i.e. for the conservation of the public health, it is necessary to improve the level of knowledge about this issue. Although there has been a lot of reporting on that issue in the media lately, the approach has been mainly sensationalistic and often scientifically unfounded. This is the reason why a national body responsible for the adaptation in the

field of health must pay attention to public education and information that are founded on relevant data.

Providing better access to health care services at a local level

To strengthen and improve the health system is a very complex and demanding process which depends on many factors, especially on social-economic development of certain regions. It is difficult to plan and ensure adequate health protection for scarcely inhabited and remote areas with a poorly developed economy and infrastructure. This is a problem that needs to be tackled together with other sectors. However, considering the potential hazardous situation of the population in these areas, it is necessary to project feasible adaptation measures in an effective and rational way. Namely, it is necessary to consider which areas are particularly vulnerable, in which temporal periods and in which weather conditions or during which natural events. In accordance with this, services could be organized that would come into action in specific situations. In these processes, cooperation with local authorities is essential.

Improving capacity to act in emergency situations

Since an upward trend of floods, fires and heat waves is expected in Serbia in the time to come, it is necessary to increase readiness of health care services to act in such situations. These activities must be synchronized and coordinated with the activities of other institutions, above all with the Center for Emergency Situations and the police in general, further with the army, the Red Cross Serbia and rescue services.

Development of early warning systems for extreme weather conditions

Although weather forecasts are available (by the Republic Hydrometeorological Service) and regularly presented in the media, it is necessary to develop better cooperation between all health institutions and other organizations and services that are relevant for the population protection in such situations.

3.3. AGRICULTURE

3.3.1. Agriculture in Serbia

Agriculture is a very important economic sector in Serbia. Agricultural production made up 9% of the total gross national income in 2011⁴¹. This is a very high percentage in comparison to the 27 EU countries, where the percentage of agricultural production lies below 3% of the gross national product⁴². Since the crisis period in the early 1990s, when agricultural production suddenly dropped, this sector has recovered gradually. In 2000, the income from agriculture increased to 21.9% of the GDP. After this period, the share of agriculture in the GDP has fallen again, but more as a result of the development of other sectors than because of a decline in agricultural production itself. Projections about the agricultural development are that the share of agriculture will continue to drop and that by 2020 it will fall under 8%⁴³.

Agricultural products make up a substantial component of exports with more than 20%. Apart from this, the number of persons employed in this sector amounts to 21%. In some regions in Serbia, agriculture represents the core activity and a large number of households depend on it.

The natural potential for the development of agricultural production is extraordinarily large. Climatic conditions are favorable; there is a good quality of soil which mostly has not yet been notably polluted. Serbia also owns considerable water resources that could be used for the development of agricultural production. More than half of the country's territory is agricultural land (4,867,000 ha), which explains such a high share of agriculture in the overall production and positively defines Serbia as an agrarian country. In addition, Serbia has a long tradition of agricultural production, which indisputably represents an important capacity.

In spite of the large potential, the current state of agriculture in Serbia is unfavorable. One of the fundamental causes is that for many years there has been no regional policy and the rural development has been neglected. With the centralization of Serbia, rural areas of the country that constitute pillars of the

agricultural production have been neglected in the development policies. Such an approach has resulted in a persistent and rapid depopulation of rural regions, impoverishment of agriculturists and weakening of their competitiveness, as well as insufficient development of infrastructure which is necessary for successful agricultural production and overall rural development.

Another problem in the development of agriculture is the instability of agrarian policy. Only in the period from 2000 to 2009, the agrarian policy changed three times. Every inconsistency in the policy caused a decrease of investments in agriculture.

Small farms prevail in Serbia. The average size per holding is about 30,000 square meters, which is below the EU27 average, and incomparably less than in countries of Western Europe. Such small farms delay the development of agricultural production, make implementation of modern production technology difficult and finally, reduce competitiveness of agricultural producers. Even 75% of agricultural households farm less than 50,000 square meters⁴⁴. There is almost no commercial production in such small-farm areas; instead, they mainly produce for their own purposes.

In such a difficult situation in general, it is evident that the state of agriculture is vulnerable, even without impact of climate change. The planning and implementation of adaptation measures is a particularly complex process that requires expertise and substantial financial means even from individual farm holders. We can say that the readiness of most farmers is not at the satisfactory level, i.e. that most of them have a very low adaptive capacity. The elevation of their adaptive capacity depends considerably from the overall rural development. Therefore, the strengthening of rural and agricultural production development represents a precondition without which implementation of adaptation measures would be hardly feasible.

3.3.2. Structure of agricultural production

With the share of about 60% in Serbia's agricultural production, arable farming is dominant, followed by animal husbandry with about 30%, pomiculture with 8% and, lastly, viniculture with almost 2%.

⁴¹ Ministry of Finance, 2011

⁴² <http://epp.eurostat.ec.europa.eu>

⁴³ Bajec et al, 2010

⁴⁴ ARCOTRASS, 2006



III.9. Overview of the harvest amounts in tons from 1947 to 2010 by major field crops. Source of information: Statistical Office of the Republic of Serbia, <http://webzrzs.stat.gov.rs>.

The most common field crop in Serbia is maize, which makes up a half of the sown area. It is followed by wheat, sunflower, soya, sugar beet, rape, and barley. In 2011, a total of about 2,200,000 ha of field crop was planted, 1,260,000 ha with maize, 490,000 ha with wheat, about 170,000 ha each with sunflower and soybeans and about 56,000 ha with sugar beets. By reviewing the production development of major field crops in the period from the World War II until today, we see that the production of maize, wheat and soybeans has grown rapidly until the mid-1980s. Then, there was a drastic decline in production in the 1990s as a result of the overall crisis situation in the country. It is evident that the agricultural production has not fully recovered ever since and that the harvest amounts are still on a low level.



III.10. Number of cattle, pigs and sheep in Serbia from 1947 to 2010. Source of information: Statistical Office of the Republic of Serbia, <http://webzrzs.stat.gov.rs>.

Pomiculture represents a significant agricultural branch in Serbia. Orchards make up almost 5% of total arable land (approx. 250 000 ha), and the most common cultures are apples, plums, sour cherries and raspberries. In the last several years, in all these cultures a relatively steady production level has been observed.

Also, the crop of basic cultures in the last decades has been marked by large fluctuations as a result of various factors (world market demand, internal agrarian policy, etc.), but one of them has originated certainly of climatic conditions.

Livestock production in Serbia has also been facing problems in the last decades. Generally, the number of cattle has been declining, for many years most strikingly in the sheep breeding. Pig breeding is most common in Serbia, but it is obvious that the current number of animals does not reach the figures from the mid-1980s.

3.3.3. Climate change impacts on agriculture

Agriculture is by nature highly dependent on climate change. Different technological solutions are being developed and implemented with the aim to reduce the dependence of agricultural production on unpredictable climatic conditions, primarily in terms of protective measures from adverse weather conditions (greenhouses, hail suppression, anti-hail net), irrigation, development of animal breeds and plant varieties more resistant to unsteady conditions, use of artificial fertilizers and chemical protective agents. Despite all this, agriculture is still very vulnerable to unsteady weather conditions, and thus to long-term climate changes.

The exposure of agricultural systems to climate change differs from region to region. Almost all projections agree that poor and developing countries, especially those in the tropical and subtropical zones, will be most affected by climate change. Developed countries in moderate climatic areas will be less affected, not only because of climate characteristics, but also because of their better adaptive possibilities. In certain regions of the world, even a production increase is expected due to rising temperature and increase of carbon dioxide concentration.

Two especially vulnerable regions represent Southern Africa and South Asia, where extremely great losses in the agricultural production is expected in the coming decades⁴⁵.

According to most models, there will be slight production increase in Europe due to climate change⁴⁶.

⁴⁵ Lobell et al, 2008

⁴⁶ IPCC, 2007

As per projections of Ewert et al (2005), the appliance of highly developed technology in the Western European countries will lead to further crop increase, which will be additionally enhanced by rising temperatures and carbon dioxide concentration. On the other hand, in the Mediterranean and Southeast Europe a decrease in the yield of grain is to be expected⁴⁷.

In the south continental agricultural region of Europe, including Serbia, it is expected that the major threat will be the temperature increase and droughts in the summer periods, which will have a negative impact on many crops⁴⁸. It is projected that some cultures will be positively affected, which will lead to better yields or an increase of sown land, in which these cultures can be cultivated.

The predictions of climate change impacts to agriculture in Serbia are not promising. Several models project a decrease in yields for almost all arable cultures. A slight increase is expected only for maize, but only in case of an intensified irrigation. For the territory of Vojvodina it is projected that the rising temperature and summer droughts will generally more affect the yields of spring sowing than winter crop. In case of winter sowing, more favorable effects are expected to exceed the indirect negative ones, due to prolonged vegetation season⁴⁹. Of course, climate change impacts will have a variety of manifestations in different areas in Serbia, but there is still no reliable information on different regional scenarios. Nevertheless, it should be mentioned that the most important arable land in Serbia lies in Vojvodina and in the valleys of large rivers, just in that lower part of the country where drier climate conditions are expected. The assumed higher frequency of natural disasters and extreme weather conditions will also affect agricultural production, primarily the plant production.

Apart from the direct climate change impact that will be reflected in temperature and precipitation regime changes, production can be also endangered by various types of diseases and parasites whose incidence and spread may alter due to climate change. The prolonged vegetation seasons due to increased winter or early spring temperatures would lead to a higher possibility of disease or insect occurrence. According to the survey results among farmers that were carried out within the

project ADAGIO⁵⁰ a larger incidence of fungal diseases has been observed with grain and fruit products over the past years. On the other hand, it is to be expected that in case of some types of insects, particularly those that depend on the soil moisture, it will come to decreased incidence and population due to prolonged dry periods.

Another adverse phenomenon that might be enhanced by climate change is erosion of soil. The expected higher frequency of floods and high waters will certainly influence intensity of water erosion in some areas. Besides, large part of Vojvodina is most susceptible to the wind erosion⁵¹ that could be significantly increased due to prolonged periods of drought and high temperatures.

3.3.4. Adaptation to climate change in the field of agriculture

The main adaptation measures that should be considered in the field of agriculture relate to the strategic level, i.e. to elaboration of strategic documents and overall agrarian policy making.

Only a small portion of Serbian budget goes to agriculture. Only 2-3 percent of the national budget has been assigned to agriculture over the past years, which is very little in comparison to the other European countries. It is still early to speak about financial impacts of climate change. As already mentioned, they could be either positive or negative, depending on our readiness to react and to adjust to them. In any case, the national spending on agriculture would need to increase if we want to adapt agricultural production adequately to the changed environmental conditions, i.e. to use potential benefits of climate change.

Strategic planning of adaptations to climate change in the field of agriculture

Serbia has still no strategic document about the long-term planning of the agricultural development. The elaboration of a Strategy of Agricultural Development has been announced for some time now, but, so far, not even a draft of such a document has been made. The elaboration of a document, i.e. a Strategy of Agriculture and Rural Development is stipulated in the Law on Agriculture and Rural Development⁵² and it should be made for a period of ten years. Bearing in mind that this

47 Ibid.

48 Iglesias et al, 2007

49 Lalic et al, 2011

50 <http://www.adagio-eu.org/>

51 Savic et al, 2002

52 „Official Gazette of the Republic of Serbia", issue 41/09

should be a key document providing guidelines for future development of agriculture, it represents a real chance for proper assessment of the climate change impacts, as well as for review of adaptation possibilities and their planning. In the previous Strategy of Agricultural Development, adopted in 2005, climate change has practically not been considered at all. The National Development Program of Agricultural Development for the period 2011-2013 just mentions that climate change will affect agricultural production, without any projections or concrete proposals about adaptive measures. For adequate integration of adaptation aspects in this Strategy, an interdisciplinary approach will be necessary as well as involvement of experts from various fields relevant for the issue of climate change.

This Strategy should also envisage elaboration of a sectoral plan for adaptation to climate change that would include more detailed adaptive measure definitions and deadlines, as well as responsible intuitions.

As already mentioned, climate changes will presumably have different impacts in different parts of Serbia. Therefore, it is necessary to pay special attention to regional specificities. It is to be expected that certain areas affected by depopulation, depression in agriculture and economy in general, will be much more vulnerable to climate change because of their reduced adaptive capacity. Therefore, it is necessary to embed the agricultural development well in the overall rural and regional development. The strategy of agriculture and adaptation plan must try to recognize the vulnerable areas in Serbia and to suggest precise adaptation measures. In these cases, they do not have to focus solely on agricultural production, but also on aspects that are of immediate significance to its development and could be affected by climate change. This refers to the road transfer infrastructure, access to knowledge and technology, but also to overall social development of regions in order to enhance their competitiveness. A limiting factor to some extent is the fact that a national strategy on adaptation has neither been adopted nor planned in the near future. This will result in that the Agriculture and Rural Development Strategy will most likely be developed and adopted before the actual umbrella document. In order to avoid possible conflicts and contradictions with other sectors, it is essential to include representatives from all other relevant sectors into the process of elaboration of the Agriculture and Rural Development Strategy. An ideal solution would

be to unify at least parts of the development process of these documents in order to achieve the much needed harmonization.

Support to scientific sector development in agriculture

Scientific research and monitoring are of essential importance in the process of agricultural production adaptation. In Serbia, there has been a significant improvement in knowledge about the climate change impacts on agriculture over the past few years. However, in order to plan a long-term development policy in an appropriate way, scientific capacity must be further developed. It is necessary to plan substantial investments in the development of scientific institutes that have been already profiled in this field, but also of regional centers that would be able to adequately monitor regional specificities. In this regard, special attention should be paid to:

- development of climatic events and climate change monitoring and forecasting;
- development and harmonization of climatic models to Serbia and its regions;
- research and characterization of conditions and choice of appropriate plant species/hybrids;
- applied research and development of new genotypes that would be adaptable to the changed environmental conditions;
- research on diseases and pests in the changed climatic conditions;
- improvement and planning of preventive activities;
- development and adjustment of production technology to new environmental conditions;
- research that would increase efficiency of agricultural production with the aim to reduce energy consumption;
- research in order to expand and increase efficiency of the irrigation network;
- consideration of breeding of species that have not yet been bred on the territory of Serbia.

Support to development of environmentally friendly agricultural production

Agricultural production substantially depends on natural ecosystems and the processes within them. Measures of protection and preservation of natural resources could alleviate adverse impacts of climate at local level, primarily by improving and maintaining high vegetation close to the farming land, raising of wind shelterbelts, preservation of wetlands, etc.

Establishing of a national body for coordination and planning of adaptation measures in the field of agriculture

Given the importance of agriculture for the overall economy of Serbia and its great vulnerability in terms of climate change, there is a need to establish a special national body that would coordinate adaptations in agriculture on a broader level. This council should also involve the scientific sector in its work.

Development of the irrigation system

Irrigation will certainly be one of the key mechanisms for adaptation, considering that in many regions a precipitation reduction and temperature increase is expected. Also, in order to optimize the benefits of the growing season prolongation and mitigation of warmer spring and autumn temperatures, it is essential to irrigate farming land. At the national level, it is therefore necessary to work out a plan to increase area under irrigation. If real effects should be achieved in rising adaptation, irrigation has to be planned in a sustainable way and must not cause problems in other sectors. The increase in areas under irrigation at any cost and without taking into account other environmental elements, especially water resources, could easily cause additional problems and challenges in adaptation to climate change.

Increasing energy efficiency in agricultural production

The demand in energy (transport, land farming, greenhouse heating, irrigation, storage and processing of products) will certainly rise due to global change. Therefore, it is very important to find strategic solutions for the rationalization of the consumption and efficient use of energy in agriculture, in order to increase the competitiveness and the adaptive capacity of farm holders.

Improving farmers' knowledge – knowledge sharing

The competitiveness of agriculturists in Serbia is on a rather low level. One of the main reasons for this is the lack of financial means for modernization of the machinery and overall production process, but also a relatively poor knowledge about the modern agricultural production of the individual farmers. Efficient adaptation to changed climatic conditions will require implementation of modern equipment which will be adjusted to the specific, local conditions. Farmers

should be stimulated to adopt new knowledge and to constantly follow new scientific and technological solutions in order to adapt their production to the changed environmental conditions. A national counseling institution has been recently established in Serbia for agriculturists and it could certainly be used for information transfer and education of farmers in terms of climate change adaptation. It should not be forgotten that Serbia has a long and rich tradition in agriculture. In spite of the current, relatively negative situation, the existing knowledge and experience of farmers should not be ignored, as it has been passed from generation to generation for a long time already. This is especially relevant to the local specifics. Therefore, it is necessary to establish two-way communication and provide decision-makers and scientists with insight into the practical experience of farmers that could be very valuable for further planning of adaptation measures. It is very important to find an effective way to inform farmers about the weather conditions. Such information is formally available, but it is questionable how many farmers actually follow the information and make use of it in their work.

Establishing mechanisms of financial assistance to farmers

The need of financial support to farmers is more than obvious considering the present situation in agriculture. Such assistance is part of a broader agriculture and national economy development strategy, so that climate change adaptations are only one planning aspect of the agrarian policy. Nevertheless, it is important that the implementation of adaptation measures to climate change, particularly the implementation of measures aiming to increase benefits of climate change, be recognized as a priority objective of certain financial mechanisms. The regional differences should be mentioned again as well as that financial measures should be foreseen and given as support to the especially vulnerable regions or agricultural branches.

Encouraging farmers to insure their land and production

Serbia has a very small percentage of insured farming land. It is estimated that only 2% of farming land owned by individual farm holders is insured. As the insurance of agricultural crops is worldwide recognized as one of important mechanisms of adaptation to climate change, it would be necessary to approach strategically to the development of this field in Serbia. By insuring

their crops, farmers would reduce their vulnerability to changeable climatic conditions and increase their adaptation capacity.

Encouraging farmers to form and join associations

An average small farming area generally reduces the competitiveness of farm holders. The need for adaptation makes the competitiveness increasingly difficult for smaller farms, since farmers are expected to change and adapt their production, as well as to invest into new technology and machinery. By merging their farm land and forming cooperatives, farmers could more easily obtain the necessary means and know-how, i.e. by merging, they are increasing their adaptive capacity.

Autonomous adaptation measures, i.e. those measures that should be applied by farm holders, must be planned in accordance with local specifics. Such measures refer to:

- choice of new breeds/species;
- change in the way of farming;
- cultivation of new plants/animals that have not yet been cultivated in the given areas;
- implementation of irrigation systems;
- change in fertilizer types and their use;
- changes in the sowing time of some cultures;
- implementation of protection measures (anti-hail net, defrost net);
- improvement in machinery;
- insurance of crops and livestock;
- establishment of cooperatives.

For planning and appropriate implementation of the above measures, it is necessary to provide individual farm holders with counseling, training in new technologies, and, of course, financial mechanism to support implementation of adaptation measures. The choice of measures greatly depends on further scientific research and climate change monitoring at local levels.

3.4. ENERGY

3.4.1. The energy sector in Serbia

The energy sector in Serbia is dominantly dependent on fossil fuels. According to the data of International Energy Agency (IEA) as well as some other international sources, Serbia is annually emitting about 60 million tons of carbon dioxide⁵³, out of which the two largest lignite-fired power plants in Kolubara and Kostolac emit about 40 million CO₂ for the electricity production. Apart from this, Serbia still has in these largest coal basins about or above 2 billion tons of confirmed reserves in lignite. There are also quantities of lignite in Kosovo⁵⁴ on the country's balance condition, approximately 2 billion tons. Therefore, in Serbia including Kosovo, there is totally 4 billion tons of lignite (and equally as much not yet into the environment emitted reserves of CO₂). Considering CO₂ cost projections on the international market limiting emission for the next ten or so years, that would mean that an amount of ca. 80 billion EUR would be needed for the emission of these quantities into the atmosphere. According to the calculations made in the public company Electric Power Industry of Serbia (EPS) and appropriate ministries in the energy

sector, it is estimated that EPS will have to pay EUR 700 million per year for their emissions.

Additionally, Serbia has around 5171MW of installed coal power plants, 2835MW of installed hydro power plants and around 350 MW in gas power plants. Serbia has practically no new, renewable sources of energy, apart from the large hydro power plants. In the country's energetic balance are also lignite-fired power plants in Kosovo, at the level of 1500MW⁵⁵.

3.4.2. Energy security and vulnerability of the supply system

Based on the research made by the International Energy Agency, over 20% of the world human population (1.4 billion) does not have access to electric energy, whereas over 40% (2.7 billion) depends on biomass fuel production⁵⁶. These figures may seem irrelevant for Serbia, but that is not so. Electric energy represents a growing problem, not because of the lack in kilowatt hours, but because a large part of Serbian society cannot afford it. Let us not forget that around

⁵³ data relates to the territory of Serbia without Kosovo

⁵⁴ http://www.lignitepower.com/index.php?option=com_content&view=article&id=163&Itemid=139&lang=en

⁵⁵ <http://www.elektroenergetika.info/te-sr.htm>

⁵⁶ http://www.worldenergyoutlook.org/docs/weo2010/weo2010_poverty.pdf

one million of Serbia's inhabitants are entitled to have social benefit payments for the basic 350kW per month and that also over 60% of households depend on using solid fuels for heating and cooking. These data are disturbing, taking into account, on one side, the quantities of energy that are being produced in Serbia and the very low productivity and inefficiency in the consumption, on the other.

Regardless the fact that the accessibility to energy of quality has not been regarded as one of the central topics of the Millennium Development Goals, this issue was incorporated in several of them, and as such should get utmost attention.⁵⁷ However, in order to eradicate the energy poverty, this objective has to be harmonized with activities foreseen to eliminate other forms of vulnerability or inappropriateness.

This refers particularly to sectors that are linked to energy production, but also for those that are influenced by different production systems, transportation and energy use. Therefore, the energy system is open to, i.e. it depends on a variety of unknown and changeable factors. The energy system is highly exposed and vulnerable to future climate change effects that are quite unpredictable in this part of Europe. Even more unpredictable is the extent of deployment of new technologies, which could, but needn't cope with the volatile climate factors. Apart from these, let us say, objective factors, the subjective one is very important in the creation of energy policy. The creation of energy policy greatly depends on current political factors and does not often relate to strategic, long-term goals.

The great problem in the future adaptation process is a relatively low level of technological development in Serbia which is neither able to appropriately support the newest technological concepts and systems in producing energy from renewable sources, nor in terms of distributing it. This is in particular the case with the implementation of smaller networks or decentralized wireless solutions that are independent of the centralized grid.

Serbian energy system is up to 30% based on water resources, but depending on the hydrological situation, it can sometimes be even more than that. From the standpoint of environmental protection and carbon

footprint reduction of Serbian energy production, it is necessary to switch to renewable energy sources. In order to do so, more energy must be derived from renewable energy sources in the final usage. This means that the percentage of lignite-burn energy, which makes currently 70% of final electricity supply, has to change. The technological limits of the currently installed coal power plants do not allow an easy switch to the use of renewable energy sources. So far, the responsible authorities have issued operating licenses for around 450 MW of wind power farms, which will theoretically make 20% of total consumption. This is, however, considering the current political and legal obstacles in deploying renewable energy sources, particularly wind energy, a hardly achievable goal. Even in case that this capacity is reached, the energy gained by burning coal will still remain in the first place in the energy production mix.

3.4.3. Climate change impacts on energy sector

In accordance with predictions regarding the reduction of surface waters in some climatic scenarios, it is clear that Serbia may face very great challenges due to water shortage. This shortage will affect the production in hydroelectricity plants, but also the usage of water for cooling and other technological processes that take place in the coal thermal stations. According to expert opinions based on the experience prior to 1990, Serbia has a very high potential in use of retention/reversible hydro power plants. Thus, Serbia can provide significant potentials for production of peak power and basic supply to the electricity system in Southeast Europe and Europe as a whole. However, these plants cause reluctance in the communities in areas where such plants are expected to be built as well as in civil society organizations dealing with nature protection.

The supply of the critically important biomass, which is expected to be very negatively affected by climate change, will be a great challenge. As households depend very much on electricity supply for heating purposes, the question is which solutions will be found for the combined impact of droughts and floods, or acidified agricultural land or forest biomass due to heavy precipitation. At the moment, Serbia possesses approximately 2,700,000 tons of biomass, 55% derives from agricultural production, while the remaining 45% is forest biomass. As per rather conservative estimates of the Ministry of Energy, in the Action Plan for Biomass

57 According to this estimate, 36 billion EUR will be needed annually for energy accessibility for clean cooking devices, at the global level.

2010-2012⁵⁸ it was estimated that Serbia has a potential of 3,400,000 tons of biomass. These estimates are based only on currently existing areas under “energy cultures” and in woods. With a change in orientation, a much more serious expansion of raw material based biomass could be achieved in Serbia. Depending on the orientation of state energy policy, biomass may play a very important role in the future. Should this be the case, there could be problems with biomass quantities, if climate conditions significantly change in Serbia. In any case, critically important decisions are presently being made on reforestation as the primary form of biomass increase in Serbia.

Apart from production problems, climate change could also cause substantial problems in energy distribution. Adverse climate conditions, especially natural disasters, could lead to cuts and defects of current conductions in the present centralized distribution system in Serbia, causing thus long-term problems in supplying certain areas with energy. Just such a situation happened in February 2012, when a cold wave hit Serbia and heavy snow caused great problems in supplying electric energy in some parts of Serbia.

Adaptations to climate change in other sectors, above all in agriculture, water management and industry, will also affect the energy sector indirectly. The way of energy consumption and supply, as well as quantities and sources of energy for these sectors will gradually change in accordance with the climatic impacts and with the adaptations that will be deployed.

3.4.4. Adaptation to climate change in the energy sector

At the national level, climate change has not yet been recognized as an important factor in the decision-making in the field of energy. However, in the Green Book of the Electric Power Industry of Serbia⁵⁹ an action plan for adaptation to climate change has been adopted. This is actually the first action plan that explicitly refers to adaptation to climate change. On the other hand, it only focuses on this company's activities and does not evaluate comprehensively the climate change effects on the energy sector and possibilities of adaptation.

One of the first steps of the energy sector in the process of adaptation must be the adoption of an energy strategy which would conform to other strategic documents in the field of adaptation to climate change. A new energy strategy is currently being prepared and one of the biggest challenges will be sectoral networking and integration of sectors that have not yet been included in the preparation of such documents. This primarily refers to the civil sector that has been one of the first to request multi-dimensional and intersectoral cooperation in formulating the strategy and particularly the action plans that would derive from it.

The energy strategy needs to be harmonized with numerous other strategic documents in areas that affect the functioning of the energy system:

- **Poverty Reduction Strategy** – poverty is one of the central problems in adopting and implementing of a sustainable strategy for adaptation to climate change, as well as for energy strategy. Due to the close relation between poverty and adaptation capacity, it is totally impossible to obtain real resources for an increase of adaptation capacity of Serbia, unless its population does not succeed in redirecting the surplus of social production by their economic and social activities. On the other hand, the energy sector is presently regarded as a purely profit-oriented economic activity and in this respect, the greatest changes have to be made. Poverty reduction should be one of the most significant indicators of success and one of central topics of energy concept change in the new approach in energy sector development. A decentralization of the energy system would guarantee poverty reduction as well as sustainable development in the energy sector.
- **Strategic documents in the field of water management** – Water management is of great importance to the functioning of the energy system, so that harmonization of sectoral plans is necessary in order to adhere to the challenges of energy production and obtaining of sufficient quantities of clean and sustainable energy by keeping the water resources intact.
- **Strategic documents for agriculture and forestry** – development of energy sector, development of renewable energy sources and decentralization of the energy system depend to a large extent on these two sectors. Aside from harmonization of the national documents, it is also necessary to

⁵⁸ NL Agency & MRE, 2010

⁵⁹ EPS, 2009

synchronize the national legislation with international documents, above all with the legislation of European Union. This refers primarily to the **EU climate and energy package**, but also to appropriate action plans.

On the strategic level, special attention should be paid to decentralization of the energy system. One way of implementing these activities would be the adoption of a regulation related to establishment of the local energy management that would be capable of closely evaluating local challenges related to climate change (in cooperation with responsible national and international institutions), and preparing exact and applicable “energetic” solutions.

Specific measures are required for the establishing of a sustainable energy system:

- **Taking advantage of the existing retentions and construction of new retentions** with the aim to establish a system of reversible hydroelectric plants is necessary in order to get a high level of regulatory electric power in Serbia. At the same time it is of utmost importance to reevaluate potential locations for small and large hydroelectric plants, with special attention to protection of the biological potential of waters which have been confirmed to be suitable for construction of energy plants.
- **Increase of biomass energy capacity.** The use of agricultural and forest waste is of particular importance. According to certain estimates, the biomass capacity could be doubled in Serbia, which amounts presently to approximately 3,400,000 tons. Of course, these measures must be very thoughtfully matched with the need for sustainable and climate-resistant agriculture and forestry.
- **Exploiting the potential of the channel system in Vojvodina** for collecting and forwarding of biomass to coal thermo power plants in Kolubara and Kostolac basins for use in furnaces, with the aim to reduce the amount of fossil fuel that is being burnt in these two largest polluters in Serbia. Naturally, this measure is highly questionable, as there are much cleaner ways to use biomass. Therefore, it is necessary to make a cost-benefit assessment of the use of biomass in such a way.
- **Development of small networks and independent networks.** This measure is very important for the repopulation of remote, undeveloped and energy poor regions of Serbia.

They are in different ways most exposed to diverse climate pressures. However, the distance to electric system is a large problem in most cases. Therefore, one of most important perspectives proves to be the development of different technologies and energy solutions that would be of manifold benefit to the local communities: benefit from the power tariffs, new jobs in the implementation and use of these solutions, partial or complete energy independence of these communities, and perhaps most importantly, the so called energy internetization, i.e. supply stability in challenging periods with unpredictable climate change impacts on the energy sector.

- **Network integration at the regional, national and international levels.** In developing international cooperation, it is necessary to agree on mutual goals based on the principles of decarbonization, energy poverty reduction, reaching of a satisfactory level of energy supply stability, but also energy security for all social classes.
- **Establishing a fund at the local, regional and national levels** that would be replenished and increased by means of savings, climate and ecology taxes collected from emitters of greenhouse gases, and a precise tax policy on technology use.
- **Measures to eliminate acidity in soil in order to increase biomass.** In addition to these measures, it is necessary to afforest great areas in Vojvodina, but also in other parts of Serbia. The afforestation strategy must be well balanced with the wood demand for thermal production.

Case study:

WWF – World Wide Fund for Nature

Campaign: Earth Hour

In 2007, confronted with serious scientific data concerning the effects of climate change, WWF Australia began to look at new methods to avoid the usual ways of tackling the climate change issue. A campaign was envisioned that would be based on hope not fear, and the idea that everyone can take personal responsibility for the future of the planet we live on – Earth Hour was born. Since its inception in 2007 in a single city - Sydney, Australia – Earth Hour has grown into the largest voluntary action for the environment in history. By the following year, 371 cities and towns in more than 35 countries around the world joined the event. The

campaign has experienced the largest growth since 2009, and in 2012 more than 6,950 cities and towns in 152 countries and territories have switched off their lights for Earth Hour, sending a powerful message for action to save the planet. Southeastern European countries also participate in Earth Hour. Since 2009 when the first Earth Hour was marked in Serbia, the number of cities, CSOs, businesses, and individuals taking part in the event has grown every year. In 2012, Earth Hour has been marked in 47 cities and towns in Serbia and WWF has organized local events in conjunction with local CSOs in Serbia. The increasing participation in activities linked to Earth Hour is indicative of the rising public consciousness concerning climate change and other environmental issues in the region. By raising awareness among all sectors and showcasing ways in which everyone can get involved in adapting to climate change, Earth Hour provides a good example of the kind of initiatives necessary to create the environment in which there is significant public will to address climate change issues.



Case study: Young Researchers of Serbia

Project: Sustainable transportation – a campaign to raise public awareness for climate change

Within the project, many street actions or events have been organized, with the aim:

- - to inform the general public about climate change and the necessity of carbon dioxide emission reduction;
- - to stress the importance of enhancing sustainable forms of transportation in urban areas;
- - to make the public aware of negative aspects of the extensive use of motorcars (air pollution, impact on health, energy consumption, climate change);
- - to promote alternative forms of transportation

(mass transit, cycling, walking);

- - to give exact recommendations to citizens regarding changes they could undertake in order to contribute to a slowdown of climate change.

In organized events/street actions, special attention has been paid to connecting the already existing regional initiatives by steering them appropriately to achieve the campaign's objectives in Serbia. There have been several promotional events within this project. In Belgrade, a street campaign has been arranged with the aim to promote sustainable transportation forms, while in order to observe the Car-Free Day and the European Mobility Week workshops and painting competitions have been organized for primary school students.



Case study: Center for Ecology and Sustainable Development (CEKOR)

Project: Railways and bikes for sustainable transportation in Subotica

The main objective of this project was to identify the shortcomings in the public transportation in Subotica; to determine measures for mobility improvement in local communities; to reduce the impact of traffic on the environment; and to enhance sustainability of the transportation infrastructure in the town and its vicinity. The project implementation consisted of organizing and preparing a survey study, organizing a round table and media campaigns within the decision-making process pertaining to the mid-term transportation planning in the town area of Subotica. The recommended measures were the construction and connection of bicycle tracks in the city, connection of these tracks with the railroad station, connection of regional railways with the city transportation, as well

as implementation of smaller busses and new lines in the public transportation. In addition, introduction of public bicycles in Subotica was recommended. Some of the recommended measures have already been implemented. Project results have been presented in the study „Railways and Bikes for Sustainable Transportation in Subotica – Alternative Solutions“⁶⁰.

As a civil society organization, CEKOR pays special attention to the energy sector and advocates the necessity of changes in this sector in the direction of decarbonization, development of alternative energy sources, increase of energy efficiency, and consumption reduction in the industry sector, households and transportation. CEKOR has organized and carried out numerous campaigns to answer the identified problems in the energy sector: a risk that Serbia might opt for the use of nuclear energy, problem of energy security, problem of sustainability of renewable sources, lack of funds, institutional and legal obstacles for decarbonization, problem of negative social impact of energy production, and problem of energy poverty.

Case study: ENVIRONMENTAL IMPROVEMENT CENTER (EIC)

Campaign: Not stupid!

Climate change and the consequences arising from it brought the EIC to initiate a campaign just before the UN Conference on Climate Change in Copenhagen, in December 2009, with the aim of pointing out the importance of this event. Representatives of the Center, with their partners in the local communities, organized a national campaign to raise awareness about climate change in 15 municipalities in Belgrade and 30 towns and municipalities throughout Serbia.

As an integral part of the campaign, the screening of the film “The Age of Stupid” and panel discussions have been organized, in order to inform the citizens about climate change and new scientific progress in this field.

60 Plestovic & Marconi, 2009



Their great attendance has evidently demonstrated that there is a need for organizing such events which, through a clear and comprehensive approach to this complex issue, enhance people to start thinking about climate change and its effects on the local environment.

Project: International Green Culture Festival GREEN FEST

The International Green Culture Festival Green Fest has emerged as one of the responses of the EIC to the lack of environmental discourse through cultural and educational activities. Held from 2010 in Belgrade, it has hosted more than 8000 visitors so far. Climate change and challenges it creates has been one of the steady topics discussed within the educational and film program of the festival.

The prize competition program for amateur and short films, peer education, lectures, workshops and exhibitions have raised the interest of students and teachers from more than 40 primary and secondary schools in Serbia, as well as film amateurs and professionals from 25 countries. With 40 workshops and lectures held, 12 exhibitions presented and over 120 films screened in the last two years, GREEN FEST has become the greatest traditional “green” event in the City of Belgrade.

3.5. CLIMATE CHANGE IMPACT ON BIODIVERSITY

3.5.1 Biodiversity in Serbia

Like other countries on the Balkan Peninsula, Serbia

is characterized by a very high genetic, species and ecosystem diversity. A variety of environmental

conditions, resulting from the geographical, geological, biogeographical, hydrological, pedological and climatic diversity, allow for the presence and survival of numerous species and their habitats in this region. Therefore, the mountainous parts of Serbia and the Balkans have been identified as one of six centers of European biodiversity⁶¹.

In Serbia, there are four of world's twelve biomes in Serbia, i.e. larger areas that are characterized by similar climatic and environmental conditions. They are steppe biome, biome of deciduous forests, biome of conifer boreal forests and biome of highland "tundra". There are different biogeographical classifications of Serbia. According to the classification used in European Union⁶², there are three biogeographical regions in Serbia: Pannonian, Continental and Alpine. The majority of the area belongs to the Pannonian region, in Vojvodina, and to the Continental region, which includes the largest part of Serbia. The Alpine region covers only a narrow border area in southwest Serbia and in Kosovo.

Around 1,200 plant communities and about 500 sub-associations divided in 59 vegetation classes have been registered in Serbia. Species diversity is also very high in Serbia. The number of vascular flora is estimated to be 3,600, which is 40% of all vascular plants in Europe. Nearly 10% of the total number of plants is endemic, i.e. raked only on the Balkan Peninsula or even in smaller areas. The diversity of fauna is also very high in this region, with around 50% of all vertebrates that inhabit Europe.

Serbia has not yet made a comprehensive overview of endangered plant and animal species at the national level according to the international criteria (IUNC Red List). Only a part of the assessment of endangered plant species has been made so far⁶³ as well as the assessment for all types of butterflies⁶⁴. According to the database of the International Union for Conservation of Nature (IUCN)⁶⁵, there are 50 species in Serbia that fell into one of the categories of endangered species at the global level.

Forests have a special significance in terms of nature protection, but also in terms of economy. They cover approximately 32% of Serbian territory (2 880 000 ha), which is below EU average, where forest area accounts for approximately 40%. Forest areas are also unevenly distributed, so that area covered by forest is much lower in Vojvodina, with only about 7% of total territory. The current condition of forests has been assessed as unfavorable, as a result of increased pressure on forests due to the difficult economic situation⁶⁶. The total forest area in Serbia has shown a positive trend in the last couple of decades, the main concern, however, causes the lack of quality, inadequate age structure and poor health condition of forests. Even 65% of the total forest areas in Serbia are coppices, 27% are high stands of natural origin, and 8% of forests area has been planted. In terms of ownership, forests in public property are slightly more represented than those privately owned and the ratio is 53% to 47%⁶⁷.

3.5.2. Use of natural resources

The share of forestry in Serbian GNI is estimated to be 0.54%, while timber industry participates with additional 2.47%. The total volume of wood mass in Serbia amounts to approximately 363 million m³, i.e. 161 m³/ha. According to the data from the Statistical Office of the Republic of Serbia, 2,603,463 cubic meters of timber was cut down in Serbia. Most of the cut woods were deciduous trees, while little more than 10% were conifers.

Hunting and fishing represent another important form of natural resources exploitation. The total area of hunting grounds amounts to slightly more than 7 000 000 ha, while there are totally around 80 000 hunters in Serbia. Predominant hunting game species are deer (110 000), wild boar (17000), rabbit (611000) and pheasant (410000)⁶⁸, although these estimated figures are unreliable, due to the lack of an adequate monitoring system of the hunting species quantity. The situation of wildlife population has been assessed as deficient⁶⁹, which is a consequence of an inconsistent hunting policy in the past. Additionally, poaching has significantly deteriorated the condition of game populations, as it was quite present in the

61 Biodiversity Strategy of the Republic of Serbia for the period 2011-2018, „Official Gazette of the Republic of Serbia“, No.13/2011

62 EEA map of biogeographical regions of Europe 2005. <http://www.eea.europa.eu/data-and-maps/figures/biogeographical-regions-europe-2005-with-national-boundaries>

63 Stevanović et al, 1999

64 Jakšić, 2003

65 www.iucnredlist.org

66 Forestry Development Strategy, „Official Gazette of the Republic of Serbia“, No.59/2006

67 Source: Statistical Office of the Republic of Serbia

68 SORS, 2008

69 National Strategy of Sustainable Use of Natural Resources, „Official Gazette of the Republic of Serbia“, No.33/2012

past due to bad control and generally poor economic situation in the country.

In Serbian rivers, lakes and canals 4807 tons of fish were caught⁷⁰. The quantity of fish has been constantly increasing in the last decade, which is largely an effect of an improved recording of catch as well as of a better organization of fishing areas management. Of the total catch quantity, the larger share goes to sport and recreational fishing (2805 t), rather than to commercial fishing (2002t). Most represented species are crucian carp with 844 tons, carp with 494 tons, and bream with 399 tons, followed by catfish, perch and silver carp, which was caught in quantity of 200 to 300 tons in 2010. The condition of fish resources is very difficult to estimate at the level of whole Serbia due to methodological deficiencies in the monitoring of the fish populations. Nevertheless, although with certain reservation, it can be concluded that the fish population varies from one fishing area to another.

The collection of herbs and edible plants, fungi and certain animal species is to a considerable extent also present in the territory of Serbia. It is estimated that approximately 4,000 pickers and collectors use these natural resources. Predominant form is the collecting of edible, medicinal and aromatic plants, above all blueberries, juniper, wild rose, lemon balm and marshmallow. In addition, mushrooms are also being extensively collected, especially porcini and chanterelles⁷¹. Of animals, most collected species are vineyard snails, green frogs and leeches.

3.5.3. Protected area system

Serbia has a relatively long tradition in protecting natural resources. The first nature reserves have been established in 1948 already, and the first larger protected area, i.e. Fruška Gora was declared a national park in 1960. The sudden increase in protected areas has happened in the period 1993-2002, when approximately 300,000 hectares have been put under protection⁷². In the last couple of years, the areas under protection show a negative trend, which is mainly the result of a reduction of the largest natural resource in Serbia, Stara Planina Nature Park, by 30,000 hectares in 2008.

The Law on Nature Protection defines seven types of protected areas, which include: strict nature reserve, special nature reserve, protected habitat, and monument of nature, landscape of outstanding features, nature park and national park.



III.11. Spatial and numeric development of protected areas in Serbia through the years (Sekulic, 2011)

There are approximately 518,200 hectares currently in Serbia under some sort of protection, i.e. around 5.86% of the total territory. This is clearly a very small percentage and far below the EU and global average of protected areas and represents as such a major problem in the nature protection sector in Serbia. The Spatial Plan of the Republic of Serbia for the period 2010-2020 foresees an increase of the protected area by 20%. Bearing in mind the present situation, i.e. the stagnation in establishing new protected areas, it will be quite a challenge to achieve this goal. Aside from the actual area, a great problem in the system of nature protection turns out to be inadequate and ineffective management of protected areas⁷³. Another adverse characteristic of protected areas is a very low percentage of areas that were primarily designated to be protected or that have strict regimes of protection. Three levels of nature protection have been defined in the Law on Nature Protection, out of which only the first level gives priority to nature protection and restricts other activities while the second and third level are very liberal. Among the protected areas in Serbia, there is a very small area in the first level, i.e. below 1% of territory under protection, while by far the largest part of protected areas is under the most liberal third protection regime.

The new Law on Nature Protection has introduced the concept of ecological network following the model of Natura 2000 ecological network of European Union.

⁷⁰ Source: Statistical Office of the Republic of Serbia

⁷¹ National Strategy of Sustainable Use of Natural Resources. „Official Gazette of the Republic of Serbia“, No.33/2012

⁷² Sekulić, 2011

⁷³ Biodiversity Strategy of the Republic of Serbia for the period 2011-2018. „Official Gazette of the Republic of Serbia“, No13/2011

In spite of the existence of a legal framework⁷⁴ the ecological network has not yet been fully efficient and it will take some time before this new concept of biodiversity protection will come into function in Serbia.

3.5.4. Climate change impact on ecosystems and biodiversity

Evidently, climate conditions have a substantial and far-reaching impact on natural systems. The living world has been developing in changeable climate conditions for millions of years already and obviously, most of the natural systems have had the capacity to adapt to such conditions. Finally, climatic variability is one of the factors that have contributed to the biodiversity through the history of evolution. The current climate change are considered adverse for ecosystems and species, because they are taking place more rapidly and expressively than the living systems find way to adapt to them.

In the fourth report of the International Panel on Climate Change, there are five basic assumptions about climate change impact on ecosystems:

1. Ecosystems may tolerate climate change to a certain extent. This elasticity or natural adaptation basically represents the scope of disturbances that an ecosystem could endure before it changes to another, unstable condition;
2. Ecosystems are exposed to various other anthropogenic pressures and climate change could accelerate these adverse effects. In some regions, however, a relief of certain anthropogenic pressures could be expected due to climate change;
3. Rapid climate change and other anthropogenic pressures could, through positive feedback, provoke new, unknown disturbances in the biosphere;
4. There is still insufficient understanding about the postponed reaction in ecosystems. In order to reduce unreliability and develop better adaptive reactions, it is necessary to improve our scientific knowledge about the transitional states of ecosystems and their functioning in the changeable climate conditions;
5. Climate change may provoke extinction of species which are essential for the functioning of certain ecosystems, which would at the same time decrease the adaptation possibilities of societies that depend on these ecosystems.

Alterations of natural systems which occurred as a consequence of climate change have been confirmed in numerous studies⁷⁵. This includes longitudinal displacements of some species and their altitudinal distribution changes, shifts of phenological phases, changes in migration and extensions of growing seasons. Thus, for instance, newest research of climate change impact on shifts in distribution of flora show that it has come to significant shifts in distribution of plants on high-mountain peaks in Europe⁷⁶. On most continental European peaks, the number of plant species has increased due to warming, while on the Mediterranean peaks the number of species decreased.

The impact of climate change on ecosystems in Serbia has not been examined in a systematic and comprehensive manner so far. General assessment of possible impacts has been given in the project that dealt with the climate change impact on biodiversity in Southeast Europe⁷⁷. According to this report, the region of Southeast Europe will face rising sea levels, more coastal erosion, changes of species' habitat and their distribution, and increasing problems caused by non-native invasive species. Also, it has been estimated that the region is vulnerable to climate change, particularly in parts of Mediterranean, where water scarcity is already present, further mountainous regions, coastal areas endangered by the rising sea levels and lowlands threatened by more frequent flooding.

The Biodiversity Strategy gives a general overview of possible climate change impacts on biodiversity and most endangered ecosystems. A relatively high vulnerability of ecosystems is also projected, since they are already disturbed, fragmented and exposed to anthropogenic pressures, which makes them even more vulnerable to climate change. High mountainous area, grasslands, forests, riverbanks, marsh and steppe habitats are considered to be most vulnerable. In the strategy, special emphasis has been given to the vulnerability of species related to the high mountainous habitats, whose appropriate ecological area will be significantly reduced due to the global warming.

Climate change has caught the attention of experts in the forestry sector and an anthology of papers has been

⁷⁴ Decree on Ecological Network, „Official Gazette of the Republic of Serbia“, issue 102/2010

⁷⁵ IPCC, 2007

⁷⁶ Pauli et al, 2012

⁷⁷ Laušević et al, 2008

published⁷⁸ which is completely dedicated to this issue. Following impacts are expected on forests in Serbia⁷⁹ (Medarević et al, 2007):

- shifting of some types of forests in relation to their latitude and altitude;
- changes of areal distribution of different types of forests and their relation to each other;
- distinction or reduction of certain forest communities;
- changes in the composition of particular plant communities;
- changes in the ratio of certain types of woods in regard to their exposure to light;
- forest communities will be more exposed to various adverse impacts;
- all above mentioned impacts will have a cumulative effect which will negatively affect the conservation of biodiversity and the prospect of a rational management of this natural resource.

Aside of aforementioned impacts, it is expected that trees will be more frequently affected by pathogenic agents⁸⁰ and parasitic insects⁸¹. Furthermore, a significant impact on biodiversity may also be caused by more frequent occurrences, such as floods, droughts, cold waves and, above all, fires. Although these occurrences are linked to natural processes and have happened regularly throughout history, they can be a serious threat to certain ecosystems and species in the present situation. In conditions of high anthropogenic pressure, impaired status of species and habitat protection, natural disasters may have fatal consequences for certain biodiversity elements, especially for narrowly distributed species and those with small population sizes.

3.5.5. Adaptation to climate change on biodiversity

The issue of climate change and its impact on natural ecosystems has been thoroughly observed in two most relevant strategic documents on nature protection, in the Biodiversity Strategy (2011) and National Programme for Environmental Protection (2010). Many activities regarding adaptation to climate change have already been listed in these documents. Thus, one of the goals mentioned in the Biodiversity Strategy is the

development of a national action plan for biodiversity and climate change. According to the Strategy, this action plan should be developed in a short period of time, in one to three years, and the institution responsible for it would be the Ministry of Environment and Spatial Planning. As in all other sectoral plans for adaptation, this action plan would also include the basic steps for further activities in the adaptation process, the development of the actual process and a decision on the responsible institutions that would implement the program. In any case, the action plan should also foresee the elaboration of a vulnerability assessment of the biodiversity on climate change in Serbia, which was also anticipated in the Biodiversity Strategy.

In the Biodiversity Strategy it is explicitly stated that there is a need to develop specific strategies for adaptation on climate change of protected areas and their management.

Specific adaptation measures on climate change in relation to biodiversity have to a large extent already been given in the Biodiversity Strategy and other relevant documents. The most important of them are mentioned below.

Development of a monitoring system for climate change impacts on biodiversity

Biodiversity monitoring is generally poorly developed in Serbia. Only a small number of species and habitats are being continuously monitored. In order to monitor changes of biodiversity as well as of climate change impacts, it is necessary to establish a standardized monitoring system that would allow gathering of comparable data in longer time intervals. Due to a large number of species and habitats, suitable indicators should be selected in order to rationalize the whole process. Apart from obtaining data, special attention should be paid to the publishing and exchange of information among stakeholders.

Development of an evaluation system for elements of biodiversity and ecosystem services

Natural ecosystems, species and habitats represent important resources as a component source of economy and support thus the whole society. Apart from being a supply source of many natural products directly used by humans (e.g. fish, deer, herbs, mushrooms), ecosystems provide many other services necessary for the functioning of human communities.

⁷⁸ Kadović & Medarević, 2007

⁷⁹ Medarević et al, 2007

⁸⁰ Karadžić, 2007

⁸¹ Mihajlović, 2007

Preservation of biodiversity by ecosystem services substantially contributes to the mitigation of climate change and adaptation to it.

Thus, for instance, natural habitats perform natural water treatment and provide reservoirs of drinking water, influence local climate characteristics and mitigate extreme events, wetland habitats have the ability to absorb larger quantities of water and thus preserve certain areas from high waters, many preserved natural areas used by people for recreation, ecosystems with a rich insect fauna provide pollination of crops, etc. The evaluation of these services is basically ignored, most probably due to the lack of evaluation mechanism for their actual economic value. As climate change could seriously degrade ecosystems and thus services that are used by the human community, it can be expected that they will become a restraining resource in some sectors and activities. Therefore, it is necessary to start duly with the assessment of their values, i.e. their importance for the functioning and development of the society. Of course, priorities in the implementation of adaptation measures will be set according to these values.

Expansion of protected areas

The need to expand protected areas has been addressed in many local documents, but they are also mentioned in international agreements that our country has signed. In the light of climate change, protected areas are regarded as one of the basic tools for impact mitigation. Natural, preserved ecosystems have a better elasticity, i.e. capability to adapt to changes. Just there is the significance of the protected areas, i.e. to preserve the adaptive potential of ecosystems within the largest possible areas by special preservation regimes. Protected areas may contribute to solving climate change issues by:

1. Mitigation

- Storage: Prevent carbon loss which is already present in vegetation and soil
- Capture-absorption: capturing of the remaining carbon dioxide from the atmosphere via natural ecosystems

2. Adaptation

- Protection: Maintain the integrity of ecosystems, mitigate local climate, reduce risks and impacts of extreme events, such as storms, floods and droughts
- Secure: Maintain basic ecosystem services which help people to cope with changes in water

supplies, fishing, health condition and agricultural production induced by climate change.

Protected areas are an essential part of the global response to climate change. They influence the causes of climate change by reducing greenhouse gas emission. Also, they help the society to face the impacts of climate change by maintaining basic ecosystem services which humans depend on. Without protected areas, the challenges would be even greater and their strengthening will give one of the most powerful natural solutions to the climate crisis.

For a further development of protected areas system, a detailed analysis of the existing system should be made and its shortcomings identified within the scope of ecosystems, species and habitats that are present on the territory of Serbia.

Development of ecological networks

If they are isolated and cut off, protected areas are much more vulnerable to changes and exposed to various pressures. Ecological networks are advantageous, as they are efficient, successful and cost-effective models for ecosystem management, with related laws and policies, institutions responsible for management, knowledge, human resources and capacities. There are possibilities to increase their networks at the level of whole regions and their joint effective management, as well as to improve ecosystem resilience to climate change and preserve vital ecosystem services.

In order to protect the nature effectively, it is necessary to provide a functional connection of ecosystems in wider areas. Ecological network is one of the nature preservation concepts that have the objective to connect and cross-link ecosystems. The process of developing ecological networks has begun in Serbia in compliance with Natura 2000 network of the European Union. Apart from the ecological networks that are focused on particular species and habitats, it is necessary to establish a general connection of all natural and semi-natural spatial units, regardless of their nature and value from the aspect of biodiversity. This refers to the preservation of greenways, waterways, parks, smaller green spatial units, and green corridors in the nature, but also in rural and urban areas.

Improvement of protected areas management

The effectiveness of the protected areas management in Serbia has been generally assessed as poor. In order to implement adequately adaptation measures, it is necessary to raise the capacity of managers of the protected areas and develop systems for an integrated management that would allow a better cooperation with all users of the protected areas and with all stakeholders. Particular attention should be paid to the development of mechanisms for improving the financing of protected areas.

Protection and improvement of forest ecosystems

Forest ecosystems are very important in the process of adaptation to climate change, so that close attention must be paid to their protection. Together with the forestry sector, programs should be developed for protection and improvement of the existing forest ecosystems, as well as plans for extension of forest areas. The protection of forests does not have to refer only to protection in the formal sense of the word, i.e. establishing of protected areas, but also a compliance of the manner and intensity of the forest utilization with biodiversity preservation needs.

Protection and improvement of water ecosystems

Water ecosystems are characterized by the abundance of wildlife, but they are also one of the most vulnerable natural ecosystems. The preservation of water ecosystems does not only contribute to the preservation of biodiversity, but it is also preserving the water resources and other natural resources that are of great importance for mankind. Apart from protections of preserved, natural ecosystems, it is necessary to develop plans for the revitalization of degraded water ecosystems. The development of a network of preserved water system would positively affect the adaptive capacities of other sectors, as agriculture and water management.

Monitoring of invasive species and action plans for the suppression of their spreading

In order to control the spreading of non-native species, it is necessary to develop a monitoring system of the phenomena and distribution of species of invasive character. The results of this monitoring should be used as a platform for planning measures to suppress spreading of these species and their adverse impact on native species, but also on agriculture, forestry and fishing.

Promotion and implementation of active protection measures

Ecosystems in Serbia are exposed to extensive anthropogenic pressures. In order to effectively preserve them, it is not sufficient to just apply passive protective measures. With the help of active protection measures, i.e. by implying various interventions in the nature, conditions for regeneration and advancement of biological systems should be granted. These measures refer to maintenance of habitats and vegetation, preservation of water regime, improvement of food base for specific species, reproduction and repopulation of species, etc.

Development of monitoring of species that are used for commercial purposes and action plan for their sustainable utilization

Many plants and animals are intensively used in commercial purposes. This refers predominantly for herbs and edible plants, fungi, edible snails, green frogs, leeches and venomous snakes that are being used for preparation of antidote. Collecting of these species may represent a serious threat in terms of their survival without a control mechanism. The threatened situation of these species' population could be significantly exacerbated by adverse effects of climate change and it is, therefore, necessary to increase the control of their utilization. Above all, it is necessary to develop a monitoring system for population status of species that have been collected, in order to plan their utilization.

Promotion of values of biodiversity - raising awareness

There is still a lack of awareness of the importance of biodiversity among the Serbian population. Above all, a close cooperation between the civil and governmental sector is essential for successful promotion of the importance of biodiversity preservation. Given the complexity of this field and the wide spectrum of interaction between the biodiversity on one side and the society and economy on the other, it is recommended that the campaign is approached in a strategic manner, i.e. to develop a communication strategy in the field of biodiversity and nature protection. One of the basic principles serving as a platform for campaigning should be the dependence of society and economy on some ecosystem services, whose specific market value could be defined.

4 National framework on climate change combat and disaster risk reduction

4.1. THE LEVEL OF KNOWLEDGE AND ACCESS TO INFORMATION IN SERBIA

4.1.1. International level

The *Intergovernmental Panel on Climate Change (IPCC)* is the leading international body in the field of climate change, founded in 1988 under the auspices of United Nations Environment Programme (UNEP) and World Meteorological Organization (WMO). IPCC is a scientific body and its purpose is to supply the decision-makers with relevant, scientific data on climate change at a global level. One of IPCC main activities is to produce reports on climate change at a global level based on a thorough analysis of all relevant scientific documents from at a global level. Four reports have been published so far, first in 1990 and the last one in 2007. The fifth report is in process of elaboration at the moment, planned to be published in 2013/2014. The reports represent the most important source of data on climate change, as they include all information obtained concerning all aspects of climate change. The reports are divided in three main parts that are produced by the appropriate working groups, and they are:

- scientific basis of climate change;
- impact, adaptation and vulnerability;
- mitigation of climate change;

Apart from these main reports, IPCC produces a large number of other documents, i.e. special reports and methodological reports, which are also a very important source of information for all those who are dealing with climate change.

Naturally, the *United Nations Framework Convention on Climate Change (UNFCCC)* is regarded as a very important source of information on climate change in general, including national reports from all countries – signatories.

Adaptation to climate change has been a focus of interest to scientists and politicians, as the IPCC reports showed that it is impossible to stop climate change and its effects in a short period of time and that it is necessary to develop special adaptation systems for the emerging climatic conditions. In the Action Plan of UNFCCC⁸², which was adopted by the countries-signatories at the conference in Bali in 2007, the adaptation became also formally one of the priorities in dealing with climate change. An enhancement of activities related to adaptation was further developed at the Conference held in Cancun in 2010, when the Cancun Adaptation Framework has been adopted by the signatory countries⁸³.

The main international initiative in the field of disaster risk reduction is the *International Strategy for Disaster Risk Reduction of United Nations (UNISDR)*. The *Hyogo Framework for Action 2005-2015*⁸⁴ is the primary document of this initiative.

Activities related to combating climate change and mitigating their effects have also been recognized as priorities in the European Union, so that the Directorate-General for Climate Action (DG Clima) was established in 2010. Since 2007 already, EU has been regulating fields of climate and energy in an integrated manner through the so-called “EU Climate and Energy Package” that includes a set of mandatory legal documents. Another important EU strategic document on climate is the “Roadmap for moving to a competitive low carbon

82 <http://unfccc.int/resource/docs/2007/cop13/eng/06a01.pdf#page=3>

83 <http://unfccc.int/resource/docs/2010/cop16/eng/07a01.pdf#page=4>

84 http://www.unisdr.org/files/1037_hyogoframeworkforactionenglish.pdf

economy in 2050⁸⁵. In the field of adaptation, the EU is also strategically guided by the document “White Paper on Adapting to Climate Change”⁸⁶.

4.1.2. Regional level

The key event inciting regional activities in combating climate change was the Sixth Ministerial Conference “Environment for Europe”, which took place in Belgrade in October 2007. The so-called Belgrade SEE Climate Change Initiative has been adopted at this conference⁸⁷. The initiative has the goal to strengthen cooperation in the region of Southeast Europe in the field of climate change and concluded an elaboration of Southeast European Climate Change Framework - Action Plan for Adaptation and establishment of a sub-regional Virtual Climate Change Center.

A Regional Climate Change Framework Action Plan (CCFAP, 2008) was developed in 2008 in cooperation with experts from Albania, Bosnia and Herzegovina, Former Yugoslav Republic of Macedonia, Montenegro, and Serbia. In the same year, the plan was accepted at the Ministerial Conference in Sarajevo by all five countries.

The Southeast European Virtual Climate Change Center⁸⁸ was established, also in 2008, within the Republic Hydrometeorological Service of Serbia in Belgrade. The main tasks of this center are strengthening of regional cooperation in the field of climate change monitoring, support of transfer of knowledge related to regional climate models, implementation of data obtained in impact studies, vulnerability assessments and adaptation possibilities, as well as an institutional and personal support for development of hydrometeorological stations in the region in order to fulfill tasks related to various international conventions.

Under the auspices of the United Nations International Strategy for Risk Reduction (UN-ISDR) and with the support of the World Bank, the Southeast European Disaster Risk Mitigation and Adaption Programme (SEDRMAP) has been initiated. This program is founded on the Hyogo Protocol, with the main purpose to reduce risks of natural disasters in this region.

4.1.3. National level

Only in the last couple of years has climate change become a serious issue to deal with in Serbia, so that our knowledge at the national level is rather limited, equally so the data available to climate change. United Nations Framework Convention on Climate Change was ratified in 2001⁸⁹ in Serbia and the Kyoto Protocol in 2007⁹⁰. The First report to the United Nations Framework Convention on Climate Change, adopted in November 2010, was the first document that dealt with climate change in Serbia in a comprehensive way. It included calculations of greenhouse gases emissions, vulnerability assessments in certain sectors and adaptation possibilities, mitigation assessments concerning climate change, further an overview of research and education in that field, the implementation status of UNFCCC stipulations and a summary of financial and technological needs. This detailed report incorporates all relevant data related to the issue of climate change in Serbia. It also included defined adaptation measures for some sectors, although it did not mention the institutions which were responsible for them, or the timeframe for their implementation.

The level of knowledge and the quality of climate change data have been significantly enhanced by the establishment of the Southeast European Virtual Climate Change Center in Belgrade. The Republic Hydrometeorological Service of Serbia⁹¹, where the center is located, has substantially improved its capacity, becoming thus a key institution for climate change monitoring and adaptation planning. Apart from the hydrometeorological office, important data are also supplied by the Environmental Protection Agency⁹² which has also dedicated a part of their work to climate change issues.

Experts from Serbia have participated in several international projects dealing with climate change and adaptation measures, obtaining thus a certain amount of data. For example, the EU funded ADAGIO project⁹³, related to adaptation in agriculture, or the project Adaptation Strategy Development for the Danube River Basin which is under the auspices of the International Commission for the Protection of the Danube River (ICPDR).

85 <http://eur-lex.europa.eu/LexUriServ/LexUriServ.do?uri=CELEX:52011DC0112:EN:NOT>

86 <http://eur-lex.europa.eu/LexUriServ/LexUriServ.do?uri=CELEX:52009DC0147:EN:NOT>

87 <http://www.unece.org/fileadmin/DAM/env/documents/2007/ece/ece.belgrade.conf.2007.20.e.pdf>

88 <http://www.seevccc.rs>

89 „Official Gazette of the Republic of Yugoslavia – International Contracts”, issue 2/97

90 „Official Gazette of the Republic of Serbia – International Contracts”, issue 88/2007.

91 <http://www.hidmet.gov.rs/>

92 <http://www.sepa.gov.rs>

93 <http://www.adagio-eu.org>

Since 2011, the Ministry of Education and Science of the Republic of Serbia has been financing the project entitled "Studying Climate Change and Its Influence on the Environment: Impacts, Adaptation and Mitigation". This is an interdisciplinary project, involving a large number of experts and scientific institutions. The initial project results are expected in 2014.

4.1.4. Local level

Activities related to climate change, particularly adaptive measures have not yet been developed at the local level in Serbia, so that the data accessibility to locally specific effects of climate change is very limited. Certain data are accessible for the territory of the Autonomous Province of Vojvodina and they are mostly related to agriculture, obtained during research undertaken by experts from the Faculty of Agriculture, University of Novi Sad and the Institute of Field and Vegetable Crops in Novi Sad.

4.2. INSTITUTIONAL ORGANIZATION IN THE FIELD OF CLIMATE CHANGE AND DISASTER RISK REDUCTION

The Ministry of Environment, Mining and Spatial Planning is the main responsible institution for the implementation of the UNFCCC and the Kyoto Protocol in Serbia. Since recently, there has been a special Directorate for Climate Change within the Department for EU Integration, International Cooperation and Project Management. Other ministries have also jurisdiction to deal with issues relevant for climate change, especially the Ministry of Agriculture, Trade, Forestry and Water Management, the Ministry of Infrastructure and Energy and the Ministry of Health. Of all mentioned ministries, only the Ministry of Infrastructure and Energy has a special directorate that is dealing with climate change. This is the Directorate for Sustainable Development and Climate Change in the Sector of Energy, which was established within the Department for Sustainable Energy, Renewed Energy Sources and Strategic Planning.

So far, Serbia has no official intersectoral body which would deal with climate change issues in a comprehensive way and take over the coordination of all other institutions.

In the field of disaster risk reduction, the coordination of all activities lies in the responsibility of the Sector for

Emergency Management within the Ministry of Interior. By its establishing in 2009, a unique protection system for emergency situations has been set up. The Sector consists of the department for prevention, department for risk management, department for civil protection, department for emergency situations, and the national training center. Responsibilities of this Sector include preventive, educational, informative, and organizational activities and actions in emergency situations aimed at protecting people's lives, health and property.

A regional center for emergency situations for Balkans and Southeast Europe is planned to be established in Serbia. It would be located in Nis, and it is expected to be brought into service in 2012.

Apart from the ministries dealing with climate change, there are also other relevant public institutions, such as the Republic Hydrometeorological Office, Environment Protection Agency, Red Cross Serbia, and many others. All of them conduct certain activities that are important in the process of adaptation to climate change, but so far, they have not been formally interlinked and coordinated at the national level.

4.3. STRATEGIES OF THE GOVERNMENT OF THE REPUBLIC OF SERBIA

Serbia has so far adopted a set of laws and bylaws, which are important for the mitigation of climate change, and which came out as obligations in the process of EU accession. On the other hand,

adaptation measures have been rather neglected in the national policy so far, and there is no national strategic document regulating this issue in an integrated way. Bearing in mind that Serbia has become a candidate

for the EU membership, it is expected that it will soon have to draw up a document in the form of a national plan for adaptation.

The national strategies developed over the past several years have generally mentioned the issue of adaptation to climate change. Thus, the National Strategy for Sustainable Development⁹⁴, National Environmental Protection Programme and Biodiversity Strategy⁹⁵ has paid special attention to climate change and framework adaptation measures have been recommended. The Strategy for Scientific and Technological Development has identified environment and climate change as one of the seven priority fields in the domain of science and technology.

In the earlier adopted sectoral documents, such as the Forestry Development Strategy⁹⁶, Energy Sector Development Strategy by 2015⁹⁷ and Agriculture Development Strategy⁹⁸, climate change was only recognized as one of several important factors, but no special attention was paid to the planning of actual measures for climate change mitigation and sectoral adaptation to changed climate conditions.

94 „Official Gazette of the Republic of Serbia“, issue 57/2008

95 „Official Gazette of the Republic of Serbia“, issue 13/2011

96 „Official Gazette of the Republic of Serbia“, issue 59/2006

97 „Official Gazette of the Republic of Serbia“, issue 35/2005

98 „Official Gazette of the Republic of Serbia“, issue 78/2005

It is indicative that one of the most recent strategic documents of the Government of Serbia, namely the Strategy for Sustainable Use of Natural Resources and Goods,⁹⁹ lacks a thorough approach to climate change, although it is obvious that natural resources have been extensively exposed to climate change.

Climate change and its impact have not been at all considered in the Public Health Strategy for the Republic of Serbia¹⁰⁰, even though the individual goals of this strategy are compatible with the general adaptation goals in the field of health care.

In the field of risk reduction, the National Strategy for Protection and Rescue in Emergency Situations¹⁰¹ represents a key document on this issue. This strategy has recognized climate change as one of the important factors that could influence the protection system in emergency situations, while disaster risk reduction has been considered as a national and local priority. Apart from strategic objectives, this document also gives an overview of all relevant institutions that have been included in the protection system in emergency situations. Another important document in this field is the Fire Protection Strategy for the period 2012-2017¹⁰².

99 „Official Gazette of the Republic of Serbia“, issue 33/2012

100 „Official Gazette of the Republic of Serbia“, issue 22/2009

101 „Official Gazette of the Republic of Serbia“, issue 86/2011

102 „Official Gazette of the Republic of Serbia“, issue 21/2012

4.4. THE ROLE OF CIVIL SOCIETY AND THE RED CROSS IN ADAPTATION TO CLIMATE CHANGE AND DISASTER RISK REDUCTION

Considering that institutional capacities for adaptation to climate change are still relatively weak in the state bodies of the Republic of Serbia, it is reasonable to expect that capacities of the civil society organizations are equally limited. It seems that the civil sector has pursued the focus of the state policy on mitigation measures of climate change, while the activities concerning promotion and planning of adaptation measures have been disregarded. Apart from limited capacities, additional problems in the civil sector are its poor inter-linkage and lack of organizational capacities, which makes it difficult for them to cooperate with the governmental sector and participate in the development of strategic documents. The Southeast European

Forum on Climate Change Adaptations¹⁰³ has the objective to strengthen capacities of the civil society organizations and their cooperation in regard to climate change adaptation. The Serbian national network “Climate Forum” has seventeen members.

The WWF as regional partner is also active in the national network in Serbia – “Climate Forum”. The WWF (World Wide Fund for Nature) is one of the world’s largest and most respected independent conservation organizations, and it has an active global network in more than 100 countries.

103 <http://www.seclimateforum.org>

WWF's mission is to stop the degradation of our planet's natural environment and to build a future in which humans live in harmony with nature, by:

- conservation of world's biological diversity,
- ensuring that the use of renewable natural resources is sustainable,
- promoting the reduction of pollution and wasteful consumption.

In order to achieve that, WWF is intensively working on mitigation of climate change and adaptation to it, primarily in preparation of specialized publications, raising of public awareness, as well as on influencing decision makers to provide mechanisms and means to combat climate change.

Some of Forum's activities are mentioned in this report to illustrate possible activities of the civil sector in the field of adaptation.

4.4.1. The Red Cross of Serbia

For some time already, Red Cross/Red Crescent Societies have dedicated their work to problems of humanitarian impacts of the increased number of natural disasters related to weather conditions and unpredictable risk of extreme meteorological events that are linked to climate change. These organizations are increasing their focus on preventive activities in terms of being prepared for disasters, developing early warning systems and programs dealing with climate change issues, rather than just reacting to actual catastrophic events. Through their work they try to integrate solutions of problems caused by climate change in the wide range of already existing activities, i.e. they try to avoid activities that are narrowly focused on just some aspects of climate change. Their objective is a reduction of vulnerability of those who are mostly affected by climate change by strengthening human capacities and mobilization of resources needed for adaptation to climate change and integration of risk management into the general planning and development activities.

The Red Cross of Serbia is a humanitarian, independent organization and the only National Society in the Republic of Serbia. On the basis of adopted international obligations, and of the provisions of the Law on Red Cross¹⁰⁴, Red Cross of Serbia (and its branches) plans, adopts and implements programs of emergency preparedness and disaster response.

The mission of the National Society is to reduce human suffering, by fulfilling tasks, such as: to provide help to affected persons in case of war conflicts, natural, ecological and other disasters, to save human lives and health and to spread knowledge about International Humanitarian Law. Further, to act preventively and educate the population in the field of health and social protection and to promote humanitarian values in the society, and in cases when needed, to provide social protection and shelter.

By fulfilling their goals and tasks, the National Society executes public authority vested by Law. By public authority, the Red Cross of Serbia and its network are recognized as an important factor providing protection and rescue in emergency situations, therefore, by the Law on Emergency Situations of the Republic of Serbia¹⁰⁵, an additional legal framework has been established in order to define the role of the Red Cross, as well as to define tasks that would be entrusted to the Red Cross of Serbia and its network. According to these provisions, Red Cross representatives, at all organizational levels, act as one of the National Crisis Headquarters' members and thus, participate closely in managing all phases of emergency preparedness and disaster response. Beside this, Red Cross is also responsible for providing first aid in terms of saving human lives, and participates in evacuation actions and implementation of different forms of shelter for the affected population (among other things, Red Cross is involved in providing facilities for temporary accommodation, offers tracing services and gives psychological support, provides food, water, hygiene products and other non-food items and appliances, proves help for the most vulnerable categories of the affected communities), and gives technical support in disaster response operations.

Red Cross of Serbia works on the following:

- **capacity building and development of own resources** according to highest international standards, improving mechanism of internal and external coordination of the whole network of Red Cross of Serbia for an effective disaster response, with the focus on fulfilling obligations vested by Law in regard to the position and role as an organization assisting governmental bodies in the field of humanitarian aid, especially in cases of disasters caused by nature or human actions.

¹⁰⁴ „Official Gazette of the Republic of Serbia”, issue 107/2005

¹⁰⁵ „Official Gazette of the Republic of Serbia”, issue 111/2009

- **implementation of measures and activities** (risk awareness-raising, as well as about the needs of mastering the knowledge and skills to undertake self-protection and self-help measures)
- **organizing trainings for broadest community categories** (and particularly, for the most sensitive/vulnerable ones) in the field of providing first aid or for undertaking basic measures in saving human lives and making family and personal preparedness plans
- **in case of disaster, Red Cross participates** in the implementation of measures rescuing human lives and gives assistance in the emergency stage of responding to disaster as well as in all others phases up to the reconstruction and rehabilitation.

In order to respond effectively in case of disaster, Red Cross has developed capacities by providing a consistent training to its human resources and by using standardized equipment in its work.

- In 129 municipal Red Cross organizations, 1.609 representatives of local authorities have been trained to respond to disasters,
- In 33 Red Cross organizations in municipalities and towns 33 district units have been established to respond to disaster and 671 unit members have received the training.
- Apart from the above mentioned, the Red Cross of Serbia has established and trained disaster response teams, specialized for specific tasks: a team for damage assessment caused by disaster on-the-spot, analysis of needs of affected people and coordination of disaster response operation (56), a mobile technical team (36), a team for water treatment, drinking water supply and creating basic hygiene conditions (21), a team for temporary accommodation and shelter (46), a rescue and shelter team in extreme winter conditions (23), and has water lifeguards.

For carrying out the assigned tasks, Red Cross has standardized equipment and works on strengthening its capacities and improving its performance.

The strength of Red Cross is reflected in:

1. having a developed structural network for disaster response,
2. having an internal normative order,
3. providing consistent personnel training according the highest international standards,

4. using uniform equipment according to the acquiring possibilities and to the role and tasks of Red Cross
5. the mutual solidarity – the greatest strength of the organization's network,
6. unitary form of managing preparation to act in disasters
7. unitary form of disaster response operation management

As for the climate change consequences, Red Cross implements preventive measures for all most endangered population categories and correspondently develops its capacities to undertake specific measures and fulfills tasks in order to respond to disasters caused by or emerged as a consequence or impact of climate change.

Special attention is being paid to elderly people, who are one of the main target groups designated as vulnerable in the context of climate change. In partnership with the Dutch non-governmental organization Yanos Public Health, the Red Cross of Serbia has developed the project "Community Care for the Elderly", which has been funded within the MATRA Programme of the Dutch Government. The project includes education of all relevant stakeholders at the local level, as well as education of trainers and volunteers for care for the elderly persons. There is also a plan to construct three-day care centers for elderly people, in Pirot, Indjija and Kragujevac.

The Red Cross of Serbia is experienced in responding to disasters that are directly or indirectly caused by the climate change effects. In the period from January 2005 to May 2012, the Red Cross network has provided their assistance and services to 17.653 families in 145 municipalities and towns that were affected by floods, torrent floods, landslides, severe weather, hail or stormy winds, extreme winter conditions, etc.



5. Recommendations

5.1. RECOMMENDATIONS FOR DECISION-MAKERS

Based on the situation analysis in individual sectors and on the vulnerability assessments, we are able to make some general recommendations for the decision-makers.

Adaptation to climate change must become a priority issue

Although there has been a significant improvement in adopting legal regulations related to climate change in Serbia over the past several years, the impression still remains that combating climate change has not been recognized as priority. This refers particularly to adaptation to climate change.

Adaptation does not refer only to damage reduction, but also to use of positive effects of climate change

Appropriate adaptation planning and its implementation could enhance competitiveness of some sectors or regions. Adaptation must become an integral part of the sustainable development planning.

The necessity of intensive intersectoral cooperation

As a complex process, adaptation requires well coordinated intersectoral cooperation. Adaptation measures in one sector often depend on other sectors (e.g. adaptation measures in agriculture depend to a large extent on water resources management and on development of infrastructure). Also, there are situations when a sector is not particularly vulnerable to climate change, but it is of utmost importance for the success of adaptation measures in another sector.

National adaptation programme (strategy)

Serbia has not yet developed a comprehensive strategic paper for adaptation to climate change, although its preparation was stipulated in the 2008 Strategy for Sustainable Development. The development of such a document should be priority, having in mind

the complexity of adaptations and the necessity of coordination of several sectors. Naturally, a multi-sectoral approach should be adopted in the elaboration of this paper that should also be harmonized with and integrated in all sectoral strategies.

National council for adaptation to climate change

Apart from the development of strategic documents, it is also necessary to develop organizational structures and capacity which would have the purpose to establish cooperation and synchronize activities among different sectors. One of the ways to do so would be the establishment of a national body or council for coordination of adaptation activities. Besides this highest national body, it would be favorable to form workings groups for particularly vulnerable sectors, too. One of the assignments of these bodies would have to be establishment and development of regional and international cooperation and integration of the national adaptation plan into regional plans. Apart from the governmental representatives and the scientific sector, it is necessary to include representatives of the civil sector in the work of this council.

Regional approach in adaptation to climate change

Considering the assessment that different regions in Serbia will be differently affected by climate change, i.e. that they will have a different adaptive capacity, administrative capacities will have to be developed at regional and local levels. Therefore, it is also necessary to consider adaptation to climate change during the development of spatial-planning documents.

Equal regional development is a precondition for an effective adaptation

Less developed regions are particularly vulnerable to climate change, considering their lesser adaptive capacity. Poorly developed infrastructure (energy sector, traffic, water supply) could have significant

effects on the local community and economy in the changed climate conditions.

Improvement of the climate change monitoring

In the past several years, there has been a noticeable development of climate change monitoring and research related to its effects in Serbia. These studies have mainly dealt with the climate and agriculture, while in the other sectors, such as public health and nature protection, there is still a lack of research monitoring. It is necessary to define specific indicators in order to make a rational and effective monitoring possible.

Development of educational capacity

Adaptation as a complex phenomenon demands a specific expertise. Educational institutions should consider including this subject into their programmes.

Raising public awareness

The population of Serbia has still a poorly developed awareness about climate change and its effects. During the planning of activities in regard to raising public awareness, one should have in mind that the most vulnerable parts of the population (e.g. population of remote, rural areas) also have a limited access to information, so that it is essential to develop information and education programmes for them. In order to achieve success in campaigning, it is above all necessary to define target groups and develop appropriate programmes for them.

Improvement of resources and capacities for risk reduction in case of natural hazards

A significant progress has been made in this field by establishing the Sector for Emergency Management.

However, the capacities of this sector are still insufficient, which proves the necessity for its further development as well as establishment of regional centers in especially vulnerable areas.

Giving priority to unstructured measures

As there are still a lot of unreliability in the assessment of the scope and effects of climate change, it is necessary to be cautious in the adaptation planning. Therefore, priority should be given to „soft“, unstructured measures. That means that in the beginning research and monitoring should be developed, to education and information, as well as optimization of use of natural resources, increasing efficiency and enhancing organizational structures, adapting production processes, etc. Substantial structural changes that demand large financial spending and time could be planned only if there is no doubt whatsoever in their effects and when there are no other reasons for their utilization apart from those related to adaptation.

Involvement of civil sector in the decision-making processes and development of strategic papers

Taking into account the complexity of the adaptation process, it is necessary to enable active participation of civil society organizations in the decision-making. Certainly by their expertise and knowledge gained through practical work on the ground, these organizations can be of great assistance, providing adequate solutions in the decision-making processes. Among civil society organizations special attention must be paid to those ones which work focuses on vulnerable sectors and groups in regard to the climate change adaptation.

5.2. RECOMMENDATIONS FOR CIVIL SOCIETY ORGANIZATIONS

Naturally, the priority activities of civil society organizations should be campaigns for public awareness raising and education. Two main target groups are decision-makers, i.e. administration sector and the general public.

Civil society organizations should take over the main role in placing the subject of climate change among priorities in the strategies and policies of future

development. Unfortunately, the current capacities of civil society organizations in this field are still rather modest and they would have first to concentrate on their own organizational development. This could be achieved by:

- establishing networks (national and international);
- development of international cooperation (participation in international projects);
- partnership development with professional

institutions and experts;

- partnership development with local communities and specific target groups;
- participation in the development of strategic and legal documents related to adaptation to climate change (e.g. preparation of national communication to the UNFCCC);

As the domain of adaptation is very broad, civil society organizations have the opportunity to specialize in specific aspects of adaptation, either in the vertical (national, regional, and local) or horizontal (sectoral) sense. In addition, they could also focus on enhancement of communication and cooperation between different levels or sectors.

In order to extend their influence over the governmental sector civil society organizations could refer to international conventions and documents, above all UNFCCC, but also to EU documents, such as “EU Climate and Energy Package” or “White Paper on Adapting to Climate Change”.

The highly specialized civil society organizations could also participate in research and monitoring of different aspects of climate change. The monitoring of many important indicators has still not been established in Serbia, which represents a great opportunity for the non-governmental sector. Activities in this field would have, naturally, to be agreed upon with the governmental and scientific institutions.

The civil society organizations could substantially help in the mitigation of adverse effects of climate change in particularly endangered areas or on vulnerable parts of the population. This could be achieved in different ways: by education, counseling, activating vulnerable parts of the population, establishing dialogue with decision-makers, supporting development of particular projects, etc.

The Red Cross is a humanitarian organization that is certainly playing a significant role in the process of adaptation and natural disaster risk reduction. Bearing in mind that the frequency of disaster events may increase in the future, the Red Cross will have to plan the strengthening of their capacity in certain regions. Especially challenging will be to deal with the increased need for solidarity in some risky regions. Following is a list of possible actions to improve the readiness of the Red Cross to act in conditions that are caused by climate change:

- further networking and strengthening of cooperation with other relevant organizations, such as the Sector for Emergency Management, Hydrometeorological Office, rescue services, public health institutes, etc.;
- assessments of regional vulnerability to climate change and natural disaster risk;
- adaptation of regional organizations according to climate change vulnerability of a particular region;
- participation in the development of strategic documents that are related to climate change;
- participation in the development of strategic documents that are related to public health by including also climate change factors in them;
- enhancement of international cooperation in the field of climate change;
- determination of particularly vulnerable groups of people and develop action plans to help these groups;
- developing a system of notification and information for given group of people;
- education about the potential impacts of climate change and adverse weather conditions on human health;
- cooperation with public health institutes and other health organizations in developing systems of monitoring of incidence of weather-related diseases and deaths.

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Annex 1: Glossary of terms

Climate

The statistics of weather over a period of time ranging from months to thousands or millions of years. The classical averaging period is 30 years, as defined by the World Meteorological Organization (WMO). Climates can be described as tropical, arid, polar etc. Characteristics of a climate are often described by seasons such as winter and summer, or the wet and dry times of year. In contrast, weather is the day-to-day experience of the climate, for example, a dry day during the rainy season.

Climate Change

A statistically significant change in measures of climate (such as temperature, precipitation, or wind) that persists for an extended period (decades or longer). The term climate change can be used to refer to climate change that results from both natural and man-made factors. However, the UNFCCC and this document uses the term to refer to the current human-induced climate change that is occurring, caused by human activities that are changing the composition of the atmosphere (e.g. through burning fossil fuels) and the land use change.

Climate-Related

Usually the term climate-related is used in reference to natural hazards to differentiate them from geophysical hazards. For example, floods, storms and drought are all climate-related. The term climate-related can be used without specifying whether such a hazard is attributed to climate change, climate variability, or simply the climate. "Hydrometeorological" is also used to mean the same thing.

Climate Change Related

Climate change related refers to phenomenon related or attributable to climate change.

Climate Variability

Variations in the state of the climate that can last from months to decades. Climate variability can result from natural and man-made process. However, this document uses the term to refer to natural processes. An example of such processes includes El Nino and La Nina.

Climate Change Adaptation

Adjustments in response to actual or expected climate change, to reduce negative impacts or take advantage of opportunities.

Climate Change Mitigation

Initiatives and measures to reduce the sources, or enhance the sinks, of greenhouse gases.

Disaster Risk Reduction

The conceptual framework of elements considered with the possibilities to minimize vulnerabilities and disaster risks throughout a society, to avoid (prevention) or to limit (mitigation and preparedness) the adverse impacts of hazards, within the broad context of sustainable development.

Seasonal Forecast

Provides a general indication of the likely character of the season over the next 3 months – specifically what the chances are that temperature or precipitation is likely to be normal, above normal and below-normal for the given place and time of year, based on conditions in the climate system. Seasonal forecasts indicate the likelihood of the general conditions for a particular season ahead and do not provide any information regarding day-to-day weather or extreme events.

Vulnerability

The conditions determined by physical, social, economic and environmental factors or processes, which increase the susceptibility of a community to the impact of hazards. For positive factors, which increase the ability of people to cope with hazards. (ISDR).

Weather

Atmospheric condition at any given time or place. It is measured in terms of such things as wind, temperature, humidity, atmospheric pressure, cloudiness, and precipitation. In most places, weather can change from hour-to-hour, day-to-day, and season-to-season.

Source: All definitions are taken from the Glossary of Terms in the Red Cross/Red Crescent Climate

Guide (RCCC, 2007, available online at [http://www.climatecentre.org/downloads/File/reports/RCRC_climateguide.pdf])

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